Human Performance Testing

A final report submitted to:

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Submitted by:

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EXECUTIVE SUMMARY

The project focused on evaluating human muscle power characteristics through the development of two systems: the bike dynamometer and the hand dynamometer. The goal was to experimentally determine the torque vs velocity and force vs velocity relationship of different muscle groups and compare the results to the expected theoretical behavior expected.

The bike dynamometer was improved to measure the brake torque applied by the cyclist as opposed to using the input torque to reduce the error caused by the flywheel's inertia. The hand dynamometer was refined from previous design by incorporating a stronger locking mechanism, and improved testing methodology to get more consistent and accurate runs. The hand dynamometer experimental testing was conducted across multiple resistance levels, and data collected and processed using LabVIEW. The resulting force vs velocity curves showed agreement with theoretical expectations, confirming the concave up response curve for human muscles power characteristics. Some sources of error were identified, including user flinching from the release mechanism; inconsistent forces applied across a run, the effects of fatigue during the higher resistance runs. However, despite these challenges the system was able to capture the expected overall trends for human muscle behavior. Our bike dynamometer was also conducted across multiple resistance levels. We were able to get good data to support the ideal curve that we expected to see. Most of the errors occurred due to fatigue felt by the rider after pedaling against a high resistance or pedaling the bike fast on low resistance runs. During low resistance runs we also saw observed a lot of shaking from the bike and the rider due to them trying to pedal as fast as possible causing slight errors in our loadcell output. However, after multiple runs, we were able to prove that both the human muscles tested in this experiment do indeed follow the expected curve behavior.

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1 INTRODUCTION

1.1 Background

When looking at the power characteristics for torque vs speed for an electrical system see get a response curve that is linear as seen in Figure 1.

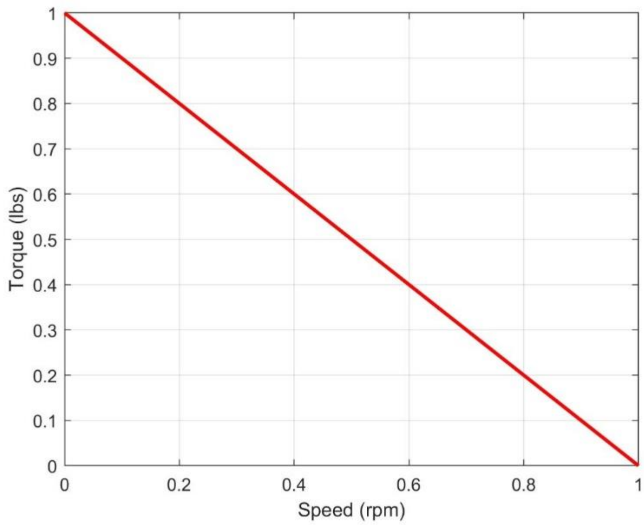


Figure 1. Electrical system power characteristics

A hydraulic system has a power characteristic for torque vs speed has as a concave down response curve as seen in Figure 2.

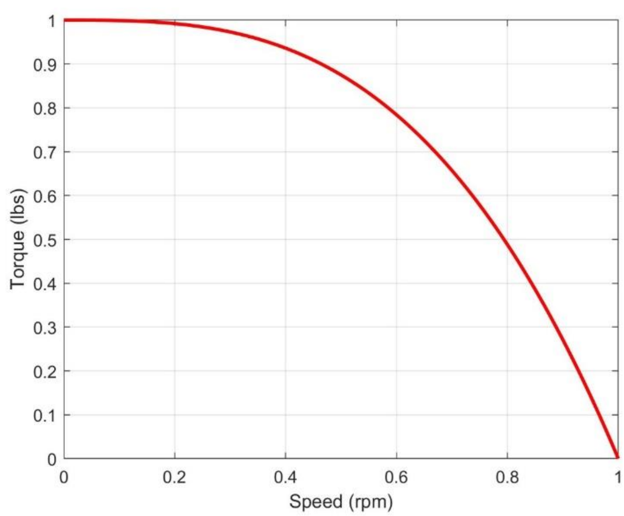


Figure 2. Hydraulic system power characteristic

Finally, when looking at a human being's muscle characteristics our theory is that the response curve for torque vs speed will be concave up as seen in Figure 3.

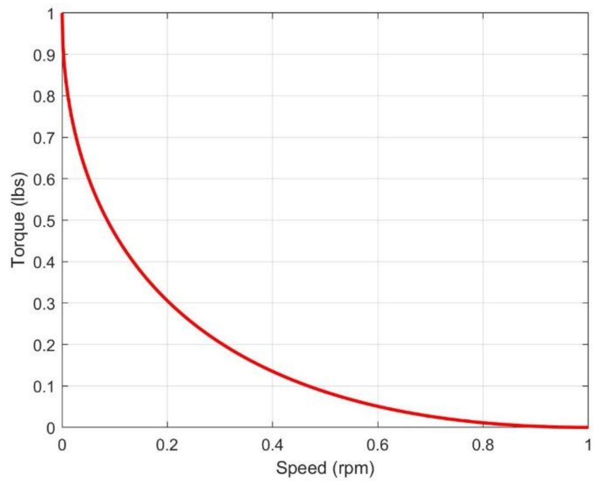


Figure 3. Human's muscle power characteristics

1.2 Previous Project Development

1.2.1 Bike Dynamometer

The previous team sensible solutions built a bike dynamometer built around the Lemond RevMaster stationary exercise bike to measure the torque and speed output of the cyclist. They implemented a torquemeter mounted on the flywheel axle that reflects the torque applied at the pedals of the bike by the cyclist. They used an IR sensor and non-reflective strips to act as spokes for the sensor to count the number of pulses that occur over a set time. The data from the sensors were then collected using an NI DAQ, and that data was complied with a LabVIEW Virtual Instrument.

However, the results of the tests showed inaccuracies in the data. As mentioned earlier, the torquemeter measures the applied torque by the cyclist at the pedals, which measures the input torque of the cyclist. When measuring input torque, it is subject to also factoring in the inertia of the flywheel as its being accelerated by the cyclist. This is seen as noise to the data that is desired for us to properly represent a human being's muscles power characteristic. To solve this problem our goal was to measure braking torque as opposed to the input torque so that the inertia of the flywheel doesn’t affect the data collected as much, giving us a more accurate representation of the muscle’s power characteristics.

1.2.2 Hand Dynamometer

The hand dynamometer was initially developed by Grayson Witkowski during the Fall 2024 semester and was later modified by the Sensible Solutions team to address the shortcomings of the original design. The device was designed to measure human grip strength by converting the applied grip force into pressure while simultaneously recording the displacement of a piston within a cylinder. These pressure and displacement signals, acquired in the form of voltage, are later converted into force and velocity for analysis.

In its initial version, the system used air as the working fluid and relied on an Arduino–Raspberry Pi setup for data acquisition. Due to the compressibility of air and limitations in the data acquisition system, the team was unable to obtain reliable and consistent results. In Spring 2025, the Sensible Solutions team improved the design by replacing air with distilled water and upgrading the data acquisition system to a NI DAQ, resulting in better measurement accuracy and data quality.

However, despite these improvements, the results were still affected by the user’s initial reaction to the sudden latch release, which introduced inconsistencies in the recorded data. The objective of the present work was therefore to implement necessary modifications and perform experimental testing to obtain data that more accurately represents actual human muscle characteristics.

1.3 Project Objective

Our objective for this project is to measure human being’s power characteristics. We will measure this using the two systems, the bike dynamometer and the hand dynamometer. For the bike dynamometer system, we will measure the muscle power characteristics when pedaling with our legs, and when pedaling with our arms to get the power characteristics of our leg muscles and arm muscles. For the hand dynamometer we are measuring a human being’s hand muscle’s grip strength power characteristics.

2 THEORY

2.1 Bike Dynamometer

The bike dynamometer which is designed around the Sunny Health & Fitness Sf-B901 chain drive stationary bike is used to measure the velocity of the flywheel and braking torque. Braking Torque can be measured by utilizing Newton's third law of reaction forces. Appendix F shows the entire process on how friction is related to what the load cell will measure.

The velocity measurement system is the same as the past group Sensible Solutions did. Non-reflective tape is placed on the face of the flywheel. An IR sensor will pulse every time it detects an edge. This will ultimately give us the velocity of the flywheel. Figure 4 shows what the flywheel looks like with the non-reflective tape.

Figure 4. Previous exercise bike's flywheel with vinyl strips

2.2 Hand Dynamometer

A hand dynamometer operates on the principle of force transmission and fluid flow. When a user applies force on the handle, the force transmitted by the lever arm to a piston within a cylinder causes a change in internal pressure and displacement of the piston. Because the system uses incompressible distilled water, the pressure generated is directly proportional to the applied force. The applied force is determined using moment balance about the pivot. A potentiometer measures the displacement of the piston, which is differentiated with respect to time to obtain velocity. Similarly, a pressure transducer converts the change in pressure into a voltage signal, which is calibrated to obtain the applied grip force. Because the raw signal contains a lot of noise, a zero-phase low-pass filter is applied to analyze the force–velocity relationship of human grip strength. The data measured by this system is then graphed against the theoretical curve defined by the equation:

$$F = F_{max}\left[1 - \left(\frac{V}{V_{max}}\right)^{n}\right]^{\frac{1}{n}} \tag{1}$$

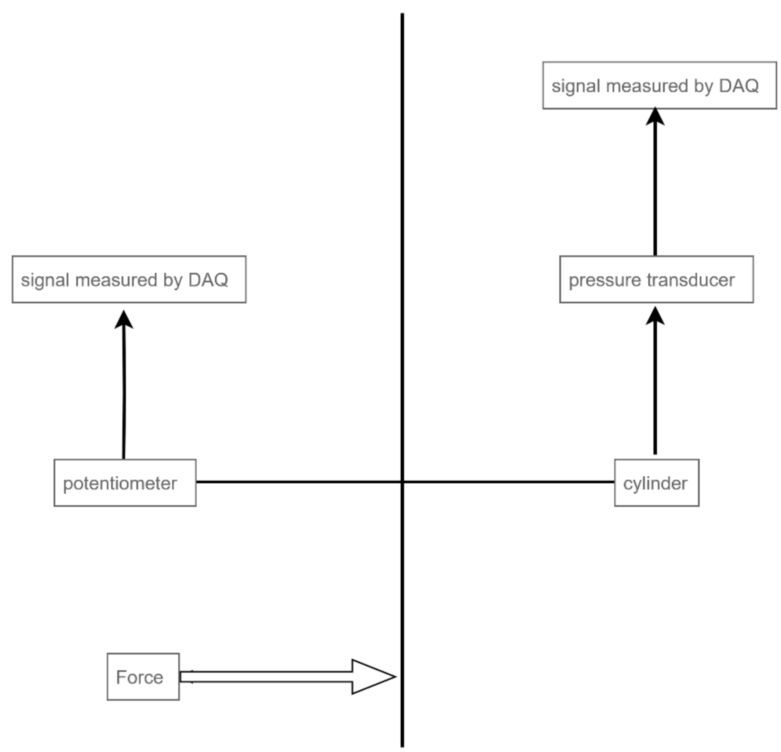


Figure 5. Hand dynamometer layout diagram

3 DESIGN

3.1 Bike Dynamometer

3.1.1 Brake Dynamometer

The exercise bike used for our setup is the Sunny Health and Fitness SF-B901 chain drive stationary bike as seen in Figure 6. Our design for the brake dynamometer consists of a brake yoke that stems from the middle of the flywheel seen in Figure 7.

Figure 6. Full bike model

The torque arms are attached to the frame of the bike using Heim joints. These joints allow the torque arm to not feel any resistance when transmitting the force from the flywheel to the load cell. The screw used to hold the brake together is what is being used to come into contact with the loadcells button.

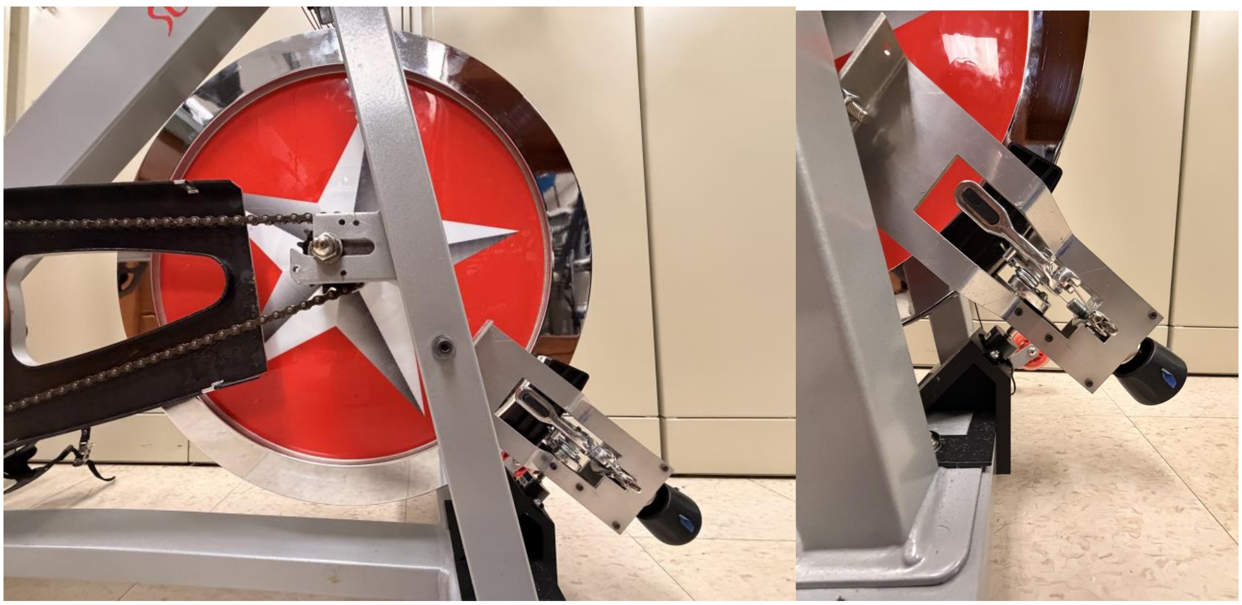


Figure 7. Flywheel with brake dynamometer(left) and close of brake system (right)

The brakes used for applying resistance to the flywheel are the brakes that come with the exercise bike, the Weinmann center pull brakes, seen in Figure 8. The knob on the brake mount allows us to change the resistance being applied to the flywheel.

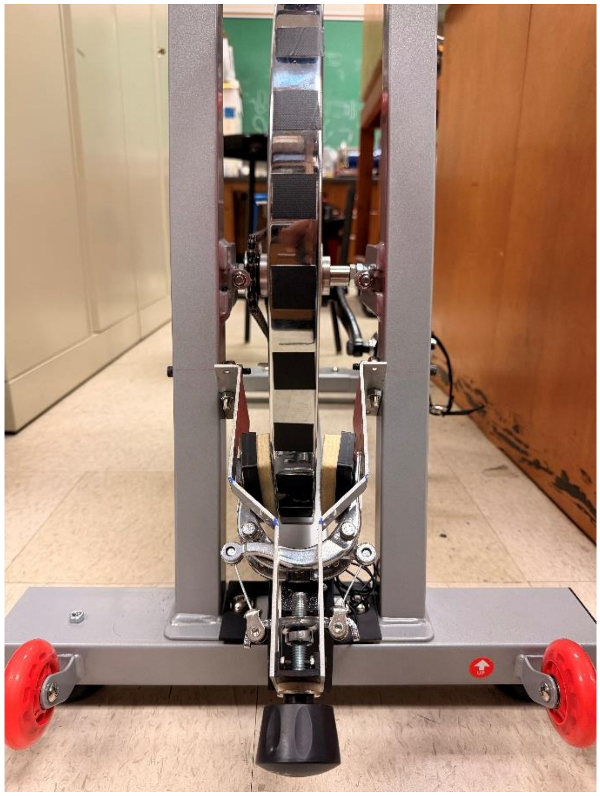


Figure 8. Brakes, adjuster knob and torque arm on bike

3.1.2 IR Sensor

The sensor that will be used for measuring the velocity of the flywheel is a reflective IR obstacle avoidance sensor module. This is the same method used by the previous Capstone team, Sensible Solutions. The chosen IR sensor was an MH-B infrared obstacle avoidance sensor from the Flying Fish sensor series. The sensor works by emitting infrared light towards the flywheel and the digital output changes based on how much light is reflected from the surface. Half of the 18.11-inch flywheel was covered with vinyl strips, and the other half was left uncovered. Each strip length was calculated using Eq. 2 where N represents the number of strips, d represents the diameter of the flywheel, and s represents the length of each strip in inches. Each strip was cut to this length and placed on the circumference of the flywheel evenly spaced.

$$s = \frac{\pi d}{2N} = \frac{\pi(18.11 \text{ in})}{2(30)} = 0.948 \text{ in} \tag{2}$$

The signal from the sensor was processed using rising edge detection. As the flywheel transitioned between the exposed surface of the flywheel to the vinyl strip, the output of the sensor changed from 0 to 1. Since one rising edge was detected for each of the 30 viny strips, 30 detected edges were one complete revolution of the flywheel. Equation 3 shows the equation used to measure revolutions per second. The complete derivation of this equation can be found in Appendix G. The time between two consecutive edges is represented by $t_n - t_{n-1}$ and multiplying by 30 and taking the reciprocal gives the estimation for the instantaneous velocity of the flywheel in revolutions per second.

$$\text{RPS} = \frac{1}{30(t_n - t_{n-1})} \tag{3}$$

3.1.3 Compressive Load Cell

Load Cell Sizing:

A TE Connectivity FC2231 compression load cell which is shown in Fig. 9 was used to measure the reaction force of the flywheel onto the brake assembly. The measurement range varied from 0-100 lbf and a range of voltage of 0.5 to 4.5 V. To size the load cell a maximum pedal force of 180lbf was assumed which is approximately the weight of the average rider. Equation 4 shows the derived equation for the friction force using the pedal length, the sprocket ratio and the flywheel geometry. The full derivation is in Appendix E, F_a is the applied force, L_1 is the pedal length, L_3 is the distance from the flywheel center to the friction force, N_{Driven} and $N_{Driving}$ are the number of teeth of the driven and driving sprockets. The calculation produced a maximum force of approximately 44 lbf therefore the 100 lbf load cell provided enough capacity while also maintaining enough sensitivity.

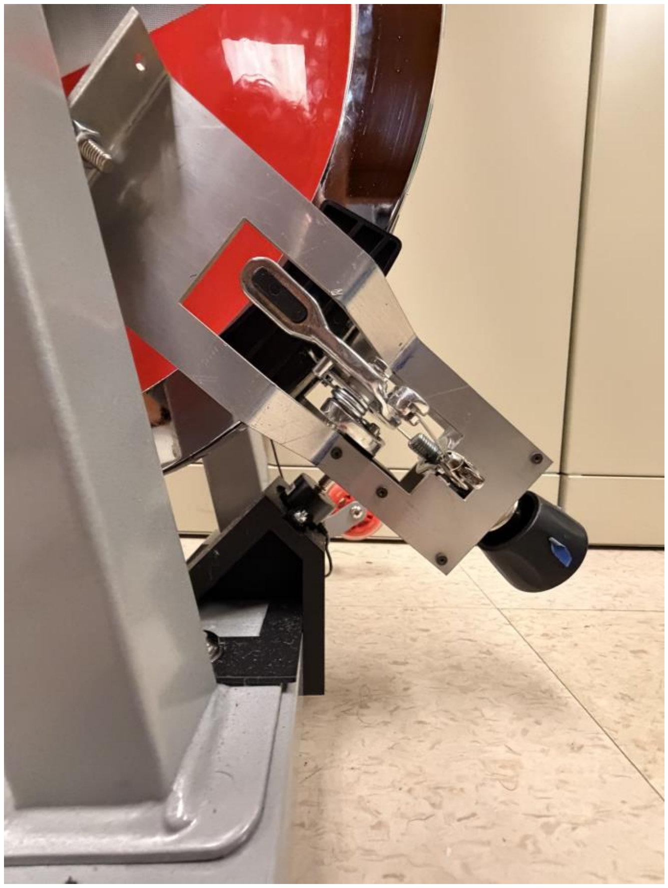


Figure 9. Torque arm resting on compressive load cell

$$F_f = \frac{F_a L_1 N_{Driven}}{L_3 N_{Driving}} \tag{4}$$

$$F_f = \frac{(180lbf)(6\frac{5}{8}in)(16)}{(8\frac{3}{8}in)(52)}$$

$$F_f = 44\ lbf$$

Brake Torque Measurement

As the bike was operated the friction between the brake pads and the flywheel assembly produced a reaction force on brake assembly which had the freedom to rotate around a rod end joint. This produced a reaction force at the point of the load cell line of action. Since the lengths of the brake assembly were known brake torque could be calculated. Since the brake assembly has a weight, which was read on the load cell as 0.56 V, this was subtracted from the load cell reading so it strictly measures the reaction force from friction. The sensor has a gain of 25 lbf/V which can be used to convert voltage from the load cell to flywheel torque. Equation 5 shows the full equation used to measure brake torque where L_7 is the load cell moment arm about the brake pivot, L_5 is the friction force moment arm about brake pivot, and L_4 is the distance from the center of the flywheel to the location of the friction force. The complete derivation of this equation can be found in Appendix F.

$$\tau_{\text{flywheel}} = (V_{\text{load}} - 0.56\ \text{V})\left(25\ \frac{\text{lbf}}{\text{V}}\right)\frac{L_7 L_4}{L_5} \tag{5}$$

3.2 Hand Dynamometer

The hand dynamometer used in this study was designed and developed by a previous senior design team, and the overall design and configuration were retained, except for the use of different, stiffer material for the locking mechanism. A stiffer locking mechanism was implemented to improve structural integrity and ensure more consistent operation during testing. Therefore, the focus was on improved data acquisition, experimental testing, and analysis.

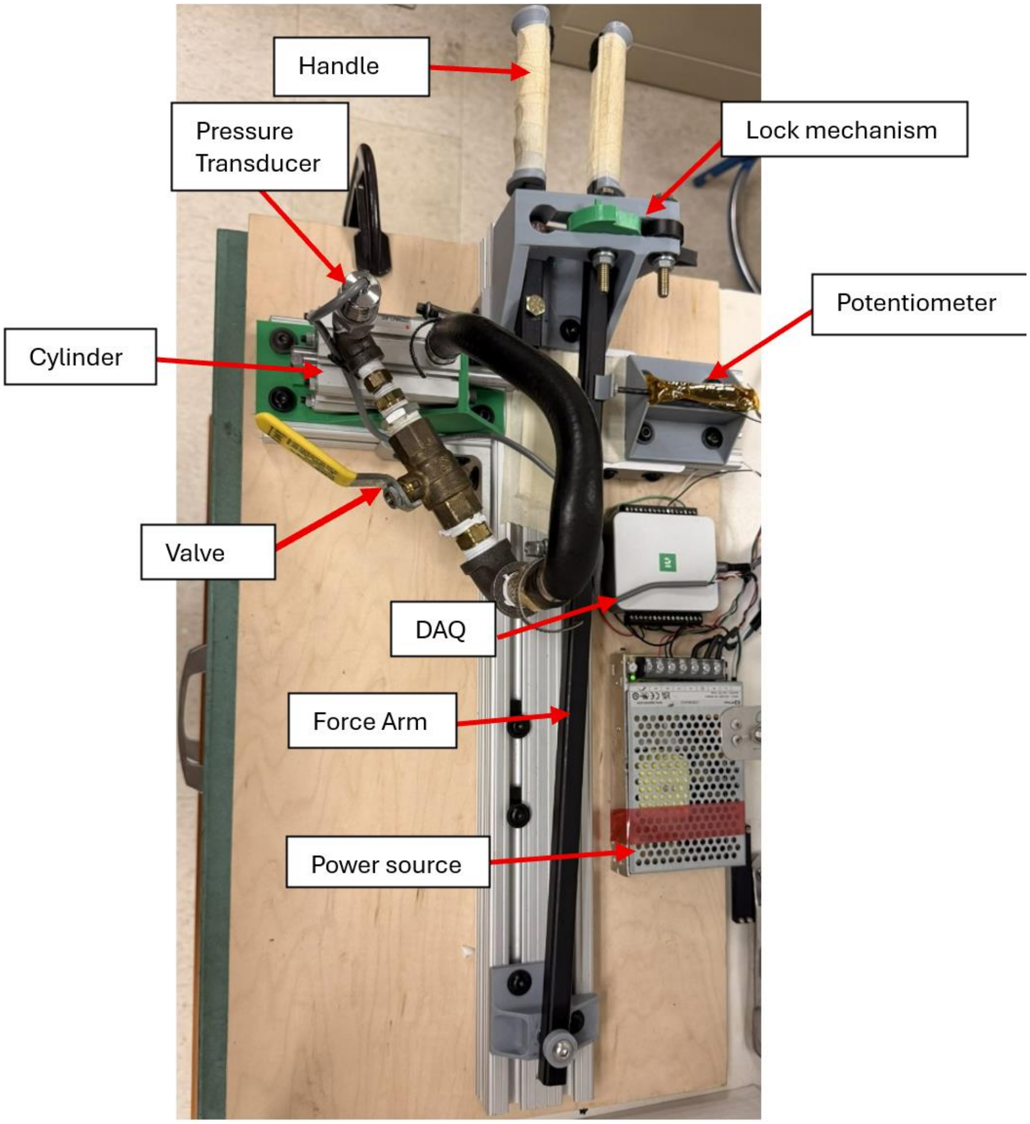


Figure 10. Hand dynamometer present configuration

4 METHODOLOGY

4.1 Bike dynamometer

The experiment test will be conducted using LabVIEW software, where we will begin by starting the program to start data collection. Then the cyclist will begin to pedal the bike and at this time we will be seeing the acceleration from the cyclist, so the program will run until we see the torque output and velocity hit a steady state for 10 seconds at this time, we will stop that test run but keep the program on and then increase the resistance applied to the bike. The same process will be repeated from no resistance to a full stalling resistance and that will be one full run for our cyclist.

We are still yet to solidify this method of testing as there are many factors such as fatigue from the cyclist having to pedal against an increasing resistance at full force in back-to-back runs. This can affect the output of the power that the cyclist is able to apply in later runs where the resistance is much higher.

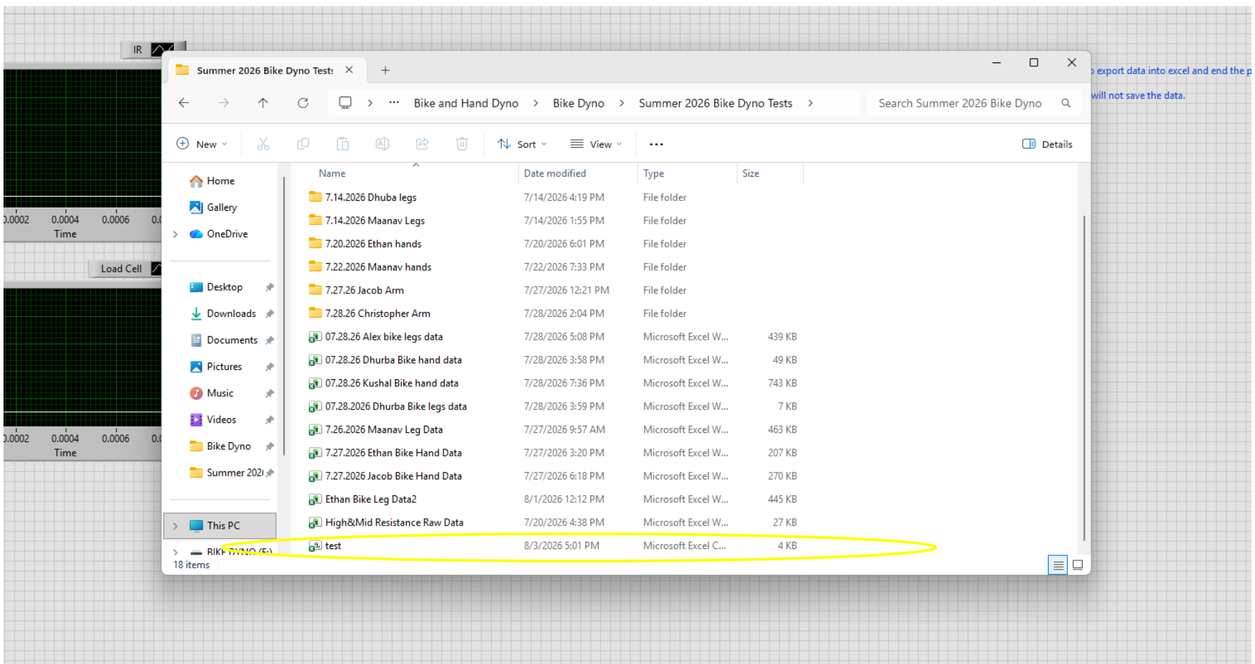


Figure 11. Bike Dynamometer Data Collection

4.2 Hand dynamometer:

The experimental test was conducted using LabVIEW software. To begin, the program was opened and the VI was executed. Initial readings for pressure and displacement were verified to ensure both the potentiometer and pressure transducer were operating to show zero displacement and zero pressure displaying the interface shown in Figure 12. While keeping the valve in a fully open position, the handle was actuated once to verify that both sensors were correctly detecting and transmitting signals.

After confirming proper functionality, the valve was adjusted to the desired resistance level. The handle was set to the fully open position, and the locking mechanism was engaged. The program was stopped and restarted to prepare for data collection. The person was asked to use an underhand grip to hold the handle with maximum strength, right after pressing the record button on the LabVIEW interface. Upon activation, the record button on the screen turned neon green, indicating that data acquisition was in progress. Without informing the subject, the locking mechanism was released, allowing the handle to close under the applied grip force.

During the process, the voltage signals displayed on the interface were monitored to ensure the readings changed appropriately. After each trial, the output file generated was checked to confirm successful data collection. The handle was then reset, the valve was set to the next resistance level, and the above procedure was repeated for high, intermediate, and low resistance settings and about 40 trials were performed.

For low-resistance trials, a sampling rate of 2000 Hz was used, and data collection occurred automatically once the record button was pressed. For medium resistance trials, a sampling rate of 1000 Hz was used. For high-resistance trials, a sampling rate of 250 Hz was used, and the record button had to be pressed a second time to terminate data recording and store the collected data.

Initial testing was carried out with multiple people across approximately ten different resistance levels. However, the data showed inconsistencies, mainly because subjects flinched during the sudden release of the latch, which affected the measurements. To improve the quality and consistency of the data collected, the approach was changed by taking about 40 trials per person in high, medium, and low resistance settings. This allowed the subjects to become more familiar with the latch release and reduced reaction-related variations, resulting in more reliable data.

Each trail produced an excel sheet dataset containing raw voltage signals. The data were trimmed to remove excess data values before latch release and after handle closure. The processed data were then used to compute force and velocity, followed by filtering and extraction of peak values. A detailed procedure is provided in Appendix A.

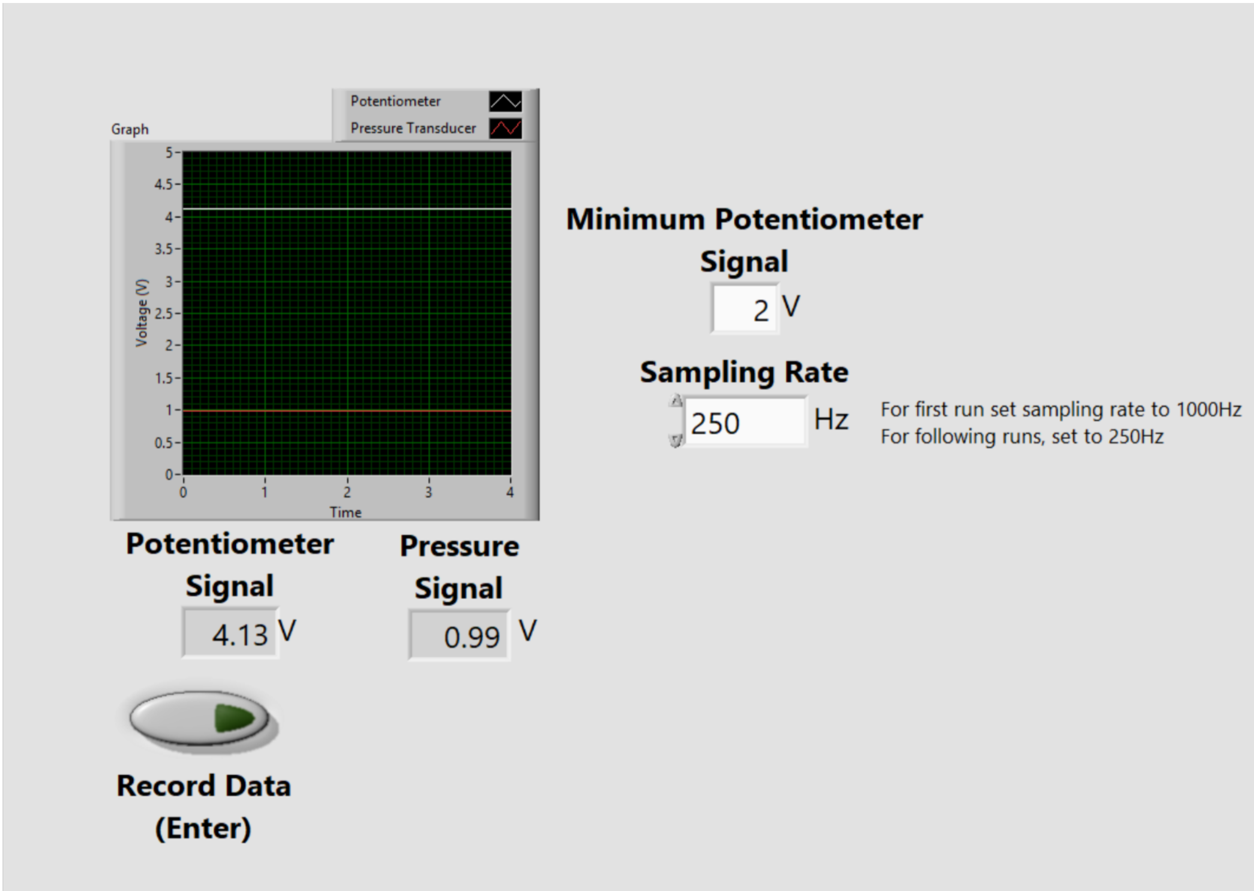


Figure 12. Hand dynamometer LabVIEW interface

5 RESULTS AND DISCUSSION

5.1 General Analysis Overview

Several sets of data were collected from both the bike and hand dynamometer and plotted against the theoretical curve given by Equation 1. To analyze the accuracy of the data and its correlation to the theoretical curve, two categories must be considered. First, how much error is contained in the measured values assuming the theoretical values are correct? Second, does the theoretical curve accurately represent the data collected? To answer the two questions, three metrics were calculated: the mean absolute error (MAE), found by Equation 6 the mean absolute percentage error (MAPE), found by Equation 7, the mean percentage error (MPE), found by Equation 8, and the R^2 value, found by Equation 9.

$$MAE = \frac{1}{N}\sum_{i=1}^{N}\left|F_{measured,i} - F_{theoretical,i}\right| \tag{6}$$

$$MAPE = \frac{100}{N}\sum_{i=1}^{N}\left|\frac{F_{measured,i} - F_{theoretical,i}}{F_{theoretical,i}}\right| \tag{7}$$

$$MPE = \frac{100}{N}\sum_{i=1}^{N}\frac{F_{measured,i} - F_{theoretical,i}}{F_{theoretical,i}} \tag{8}$$

$$R^2 = 1 - \frac{\sum(F_{measured} - F_{theoretical})^2}{\sum(F_{measured} - \bar{F}_{measured})^2} \tag{9}$$

5.2 Hand Dynamometer Analysis

From the data collected two main Force vs. Velocity graphs were created as shown in Figures 13-14.

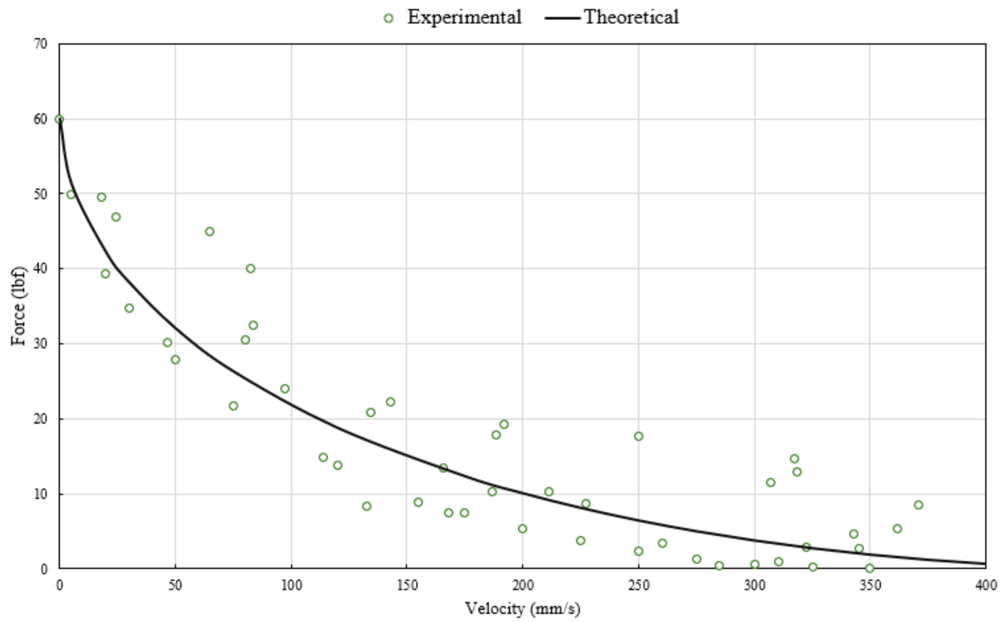


Figure 13. Force vs. Velocity Graph of Participant 1

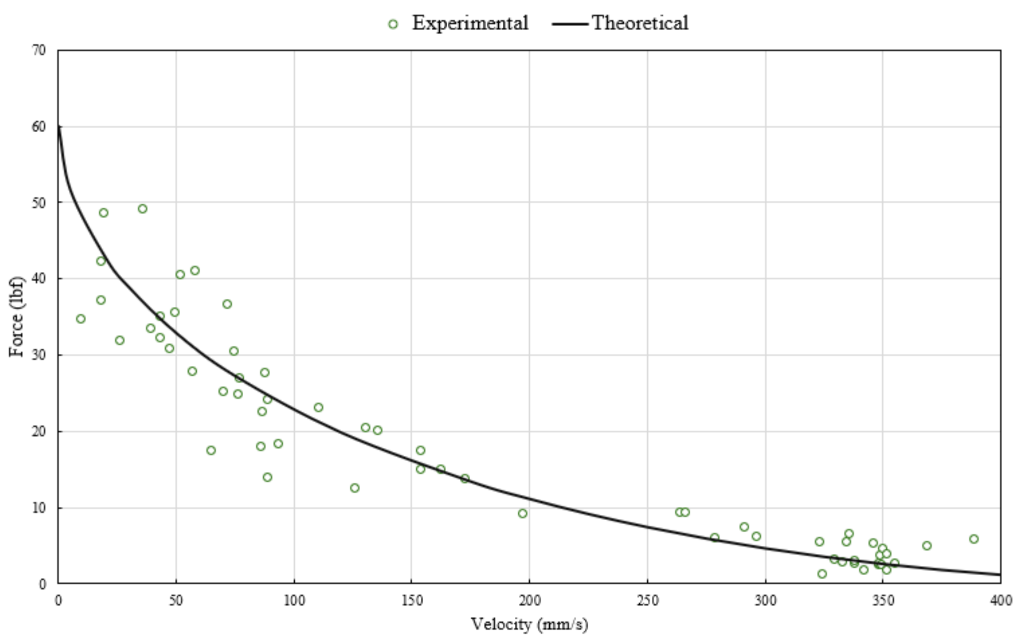


Figure 14. Force vs. Velocity Graph of Participant 2

The calculated values of the MPE, MAE, MAPE, and R^2 are shown in Table 1 below.

Table 1. Analysis Value Results

Participant	N (data points)	MAE (lbf)	MAPE (%)	MPE (%)	R^2
P1	33	3.462	34.8%	4.1%	0.899
P2	38	2.975	96.4%	48.0%	0.707
P3	60	3.229	30.5%	14.5%	0.886
P4	39	2.412	87.7%	47.3%	0.816
P5	47	5.049	60.8%	12.4%	0.852
P6	33	3.317	35.3%	12.9%	0.957
P7	34	2.923	44.5%	17.8%	0.955
P8	31	2.315	35.7%	10.9%	0.909
Average	**39.4**	**3.210**	**53.2%**	**21.0%**	**0.873**

Table 2. n Considerations (P = Participant)

n	P1 R^2 (%)	P2 R^2 (%)	P3 R^2 (%)	P4 R^2 (%)	P5 R^2 (%)	P6 R^2 (%)	P7 R^2 (%)	P8 R^2 (%)	Average R^2 (%)
0.60	81.7	59.5	82.1	71.4	80.9	87.5	92.5	82.3	**79.7**
0.59	84.2	62.9	84.2	74.4	82.4	90.0	93.7	84.9	**82.1**
0.58	86.2	65.7	85.9	76.9	83.5	92.1	94.6	87.1	**84.0**
0.57	87.9	67.9	87.2	78.9	84.4	93.7	95.2	88.8	**85.5**
0.56	89.1	69.6	88.1	80.5	85.0	95.0	95.5	90.1	**86.6**
0.55	89.9	70.7	88.6	81.6	85.2	95.7	95.5	90.9	**87.3**
0.54	90.3	71.1	88.6	82.2	85.0	96.0	95.1	91.2	**87.4**
0.53	90.1	71.0	88.2	82.3	84.5	95.8	94.3	91.0	**87.2**
0.52	89.5	70.4	87.2	81.9	83.7	95.1	93.1	90.2	**86.4**
0.51	88.5	69.1	85.8	81.0	82.5	93.9	91.5	88.8	**85.1**
0.50	86.9	67.2	83.7	79.6	80.8	92.2	89.3	86.8	**83.3**
0.49	84.8	64.7	81.1	77.8	78.8	90.0	86.8	84.2	**81.0**
0.48	82.2	61.6	78.0	75.4	76.4	87.1	83.6	80.9	**78.2**
0.47	79.0	58.0	74.2	72.5	73.6	83.8	80.0	77.0	**74.8**
0.46	75.4	53.8	69.8	69.1	70.4	79.8	75.8	72.5	**70.8**
0.45	71.2	49.1	64.7	65.2	66.8	75.4	71.0	67.2	**66.3**

0.44	66.4	43.9	59.1	60.9	62.8	70.3	65.6	61.3	**61.3**
0.43	61.2	38.1	52.7	56.0	58.4	64.7	59.5	54.8	**55.7**
0.42	55.3	31.9	45.8	50.8	53.6	58.5	52.9	47.6	**49.6**
0.41	49.0	25.3	38.2	45.1	48.5	51.8	45.6	39.8	**42.9**
0.40	42.2	18.2	29.9	39.1	42.9	44.6	37.7	31.3	**35.7**
Best n	**0.54**	**0.54**	**0.54**	**0.53**	**0.55**	**0.54**	**0.56**	**0.54**	
Max R^2 (%)	**90.3**	**71.1**	**88.6**	**82.3**	**85.2**	**96.0**	**95.5**	**91.2**	

As seen from Table 1, across all participants, the coefficient of determination averaged ($R^2 = 0.873$), indicating that the fitted data captures the overall shape of the force-velocity decay well. The mean absolute error (MAE) of 3.21 lbf across the group is relatively low. However, the mean absolute percentage error (MAPE) of 53.2% and mean percentage error (MPE) of 21% were both substantially higher than the absolute error.

The disagreement between the absolute-error metrics (R^2, MAE) and the percentage error metrics (MAPE, MPE) is the central feature of the dataset. MAPE weights each residual by the magnitude of the individual measurement rather than the size of the dataset as a whole. Because the damping force decays to near zero at high velocities, any fixed instrument error becomes a large percentage of the small force reading, amplifying the MAPE. This explains why P2 and P4—the two participants with the lowest MAE values in the group—also show the highest MAPE values, despite having among the lowest absolute errors in the sample. This suggests that the percentage error metric simply amplifies noise in the low-force tail region of the curve.

A positive MPE indicates a bias in the measurement. Because MPE retains the sign of each error, a consistently positive value across all eight participants means that the measured force exceeded the theoretical prediction more often than not, rather than the error being symmetric and cancelling out. If the errors were purely due to random instrument noise, the MPE would be expected to hover around zero. This suggests that the model slightly underpredicts the damping force, particularly in the low-velocity range where the curve is most sensitive to (n).

(R^2) is a trend-fit metric. From the table, it can be clearly seen that participants P6 and P7 achieved the highest (R^2) values (0.957 and 0.955) in the group despite having MAPE values of 35% and 44%, respectively. This is because (R^2) is only sensitive to the overall shape of the curve and not the magnitude of individual residuals. Conversely, P2's low (R^2) value (0.707) reflects a poorer trend fit, likely due to the smaller force range. Also, from Table 2, it can be clearly seen that the optimum value of (n) for nearly every participant falls within the range of 0.53–0.55, despite significant differences in their Fmax and Vmax values. While Fmax and Vmax scale the curve according to each participant's individual strength and speed, the parameter (n) determines the shape of the curve. Since the coefficient of determination reaches its highest value at (n = 0.54), this value can be selected as the optimal value of (n).

These results illustrate why a single error metric is insufficient for assessing an instrumentation system. MAE and (R^2) suggest that all eight datasets performed comparably well, while MAPE and MPE suggest that some participants performed poorly. Overall, reporting all four metrics together provides a more accurate representation: the collected data fit the overall trend adequately for every participant, while there is also a modest, consistent bias that likely occurred due to resolution limitations near the zero-force/high-velocity region rather than random measurement error.

5.3 Bike Dynamometer Leg Analysis

From the data collected from the bike dynamometer, three graphs were generated as shown in Figures 15-17. Note that each graph will show the curve fit that most closely corresponds to the data set, and that the participants are different than the participants in the hand dynamometer.

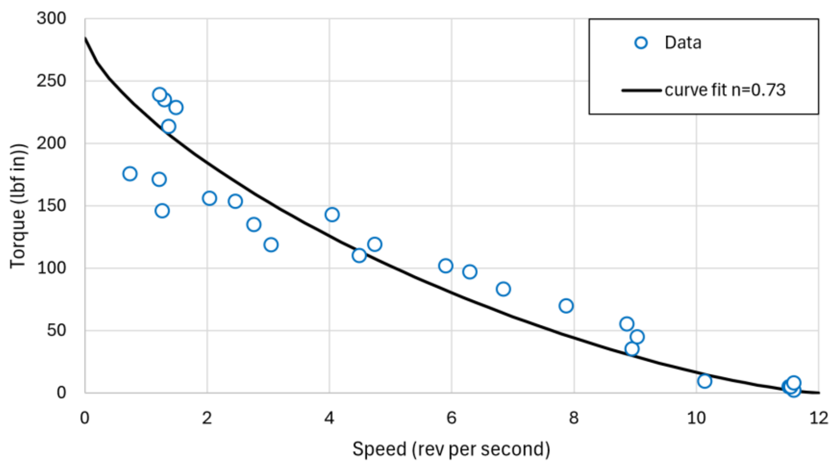


Figure 15. Torque vs. Speed Graph of Participant 1

For Participant 1, the maximum speed was set to 12 rev/s and the maximum torque was tested to be 284 lbf*in.

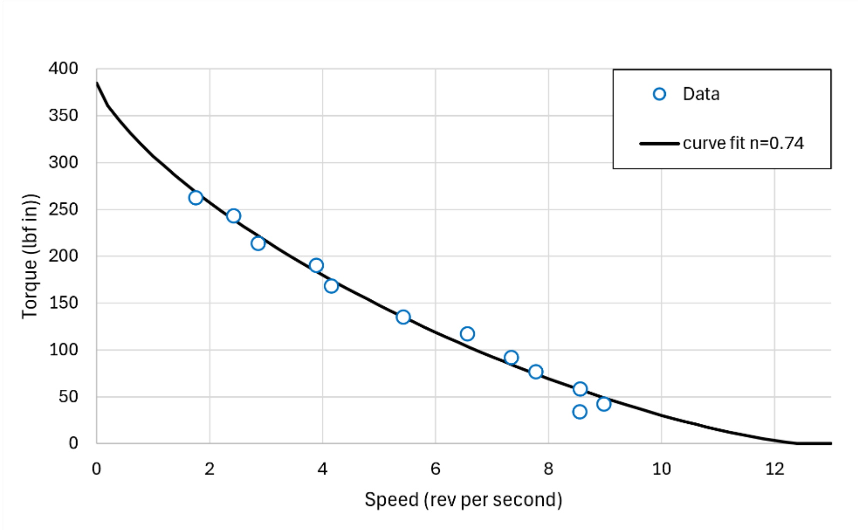


Figure 16. Torque vs. Speed Graph of Participant 2

For Participant 2, the maximum speed was set to 12.5 rev/s and the maximum torque was tested to be 385 lbf*in.

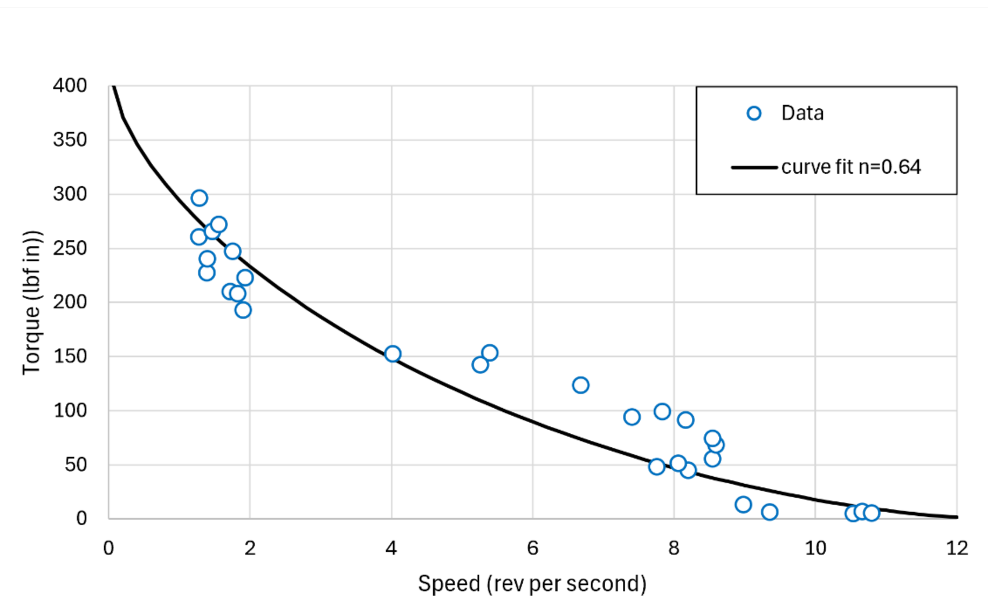


Figure 17. Torque vs. Speed Graph of Participant 3

For Participant 3, the maximum speed was set to 12.5 rev/s and the maximum torque was tested to be 416 lbf*in.

For the analysis of which value of n produced the best curve fit Table 3 shows all the considered values according to Equation 7.

Table 3. Bike Leg n Considerations

n	P1 R^2 (%)	P2 R^2 (%)	P3 R^2 (%)
1.00	46.666	3.246	-15.706
0.99	48.912	9.215	-11.169
0.98	51.131	15.071	-6.672
0.97	53.321	20.808	-2.215
0.96	55.478	26.417	2.196
0.95	57.602	31.892	6.560
0.94	59.687	37.222	10.872
0.93	61.733	42.401	15.129
0.92	63.734	47.419	19.328
0.91	65.689	52.267	23.464
0.90	67.593	56.936	27.533
0.89	69.443	61.415	31.532
0.88	71.235	65.696	35.456
0.87	72.964	69.767	39.301
0.86	74.626	73.617	43.060
0.85	76.216	77.238	46.730
0.84	77.730	80.616	50.306
0.83	79.163	83.741	53.780
0.82	80.508	86.601	57.149
0.81	81.761	89.184	60.405
0.80	82.915	91.478	63.543
0.79	83.963	93.471	66.555
0.78	84.900	95.151	69.434
0.77	85.718	96.504	72.173
0.76	86.410	97.517	74.765
0.75	86.968	98.179	77.200
0.74	87.384	98.475	79.471
0.73	87.649	98.393	81.568
0.72	87.755	97.919	83.482
0.71	87.693	97.042	85.203
0.70	87.452	95.747	86.721
0.69	87.024	94.022	88.024
0.68	86.397	91.856	89.101
0.67	85.560	89.236	89.940
0.66	84.503	86.150	90.528

0.65	83.214	82.588	90.853
0.64	81.680	78.539	90.901
0.63	79.891	73.994	90.657
0.62	77.833	68.945	90.108
0.61	75.493	63.384	89.237
0.60	72.860	57.305	88.030
0.59	69.919	50.704	86.470
0.58	66.658	43.578	84.542
0.57	63.065	35.926	82.229
0.56	59.127	27.749	79.515
0.55	54.831	19.052	76.383
0.54	50.166	9.840	72.819
0.53	45.121	0.122	68.805
0.52	39.687	-10.087	64.328
0.51	33.856	-20.773	59.375
0.50	27.619	-31.915	53.934

From the best fit curve of the specified n values, Table 4 shows the mean absolute error and the mean percentage error as well as the R^2 value.

Table 4. Analysis of Bike Leg Data

Participant	n	MAE (lbf*in)	MAPE (%)	MPE (%)	R^2 (%)
1	0.72	20.72	27.47	10.84	87.755
2	0.74	7.08	10.45	-5.71	98.475
3	0.64	23.03	43.79	-17.04	90.901

From Table 4, it can be calculated that for the data set the overall mean value of n is 0.7. The data set contains some errors, with participant three containing the most. From Figures 15 and 17, the curve of the data should be exponential as a straight line would not accurately reflect the data. This conclusion is also supported by the data collected.

For Participants Two and Three, the calculated value of MPE in Table 4 is negative which means the measured values of torque fell below the theoretical values. However, the inverse is true for Participant One, and this means that it cannot be determined whether the current range of values for n over or underpredicts the true values.

The value of R^2 for Participant Two is enough to show a strong correlation to the line of best fit and the measured values; however, due to the lack of data points in the plot it is insufficient to prove the theoretical curve by itself. Participant One's and Two's R^2 value being about 90% show that the line of best fit closely matches the measured data with some error.

The main conclusion that is drawn from the analysis of this data set, is that the curve of the line of best fit is exponential as well as the value of n is at least 0.1 higher than the average n value of the muscle characteristics used to grip calculated by the hand dynamometer.

5.4 Bike Dynamometer Arm Analysis

From the data collected from the bike dynamometer, three graphs were generated as shown in Figures 18-21. Note that each graph will show the curve fit that most closely corresponds to the data set, and that the participants are different than the participants in the hand dynamometer.

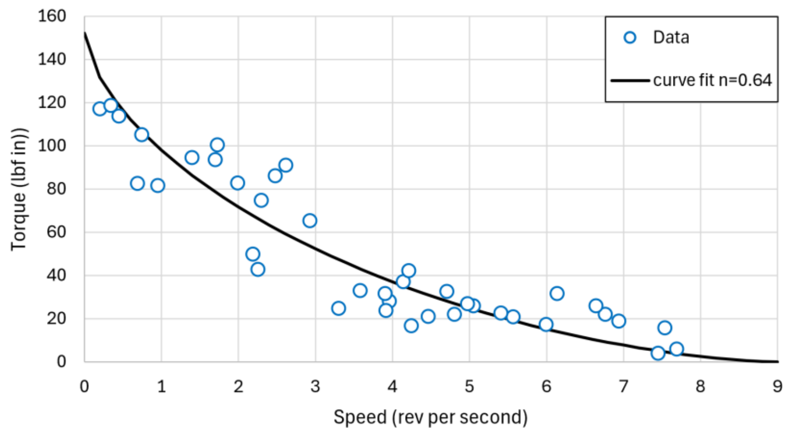


Figure 18. Torque vs. Speed Graph of Participant 1

For Participant 1, the maximum speed was set to 9 rev/s and the maximum torque was tested to be 152 lbf*in.

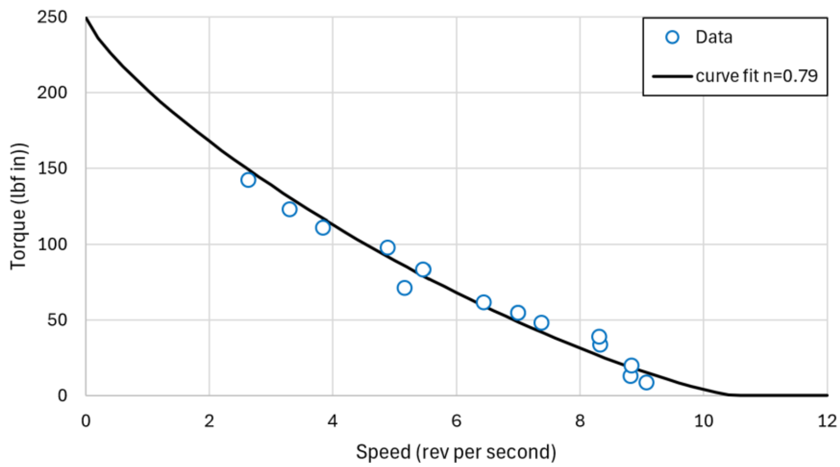


Figure 19. Torque vs. Speed Graph of Participant 2

For Participant 2, the maximum speed was set to 10.5 rev/s and the maximum torque was tested to be 250 lbf*in.

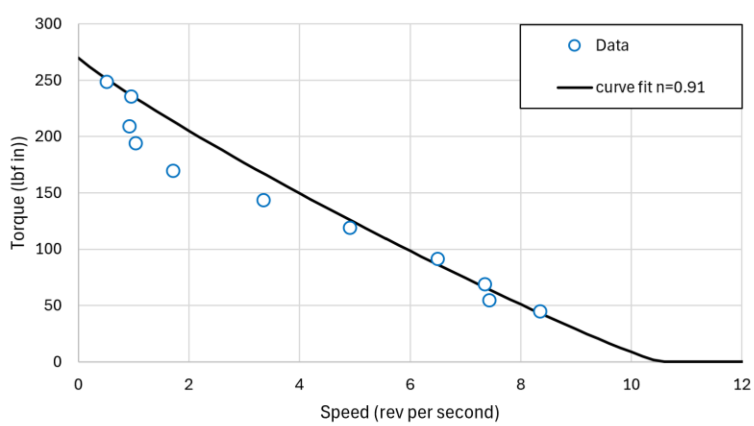


Figure 20. Torque vs. Speed Graph of Participant 4

For Participant 4, the maximum speed was set to 10.5 rev/s and the maximum torque was tested to be 270 lbf*in.

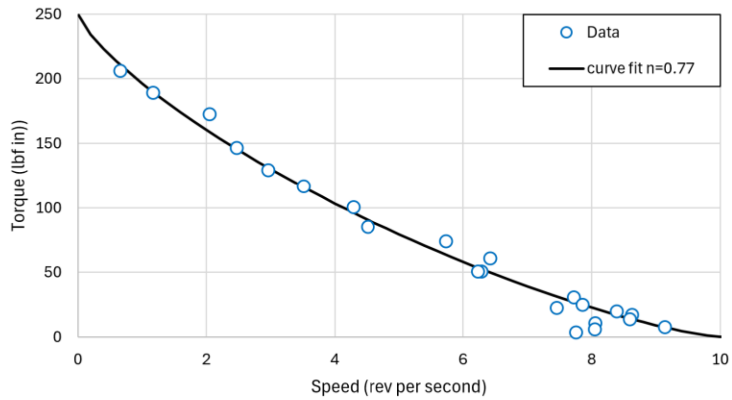


Figure 21. Torque vs. Speed Graph of Participant 5

For Participant 5, the maximum speed was set to 10 rev/s and the maximum torque was tested to be 250 lbf*in.

For the analysis of which value of n produced the best curve fit Table 5 shows all the considered values according to Equation 7.

Table 5. Bike Arm n Considerations

n	P1 R^2 (%)	P2 R^2 (%)	P4 R^2 (%)	P5 R^2 (%)
1.00	-47.640	20.172	82.712	66.537
0.99	-42.138	26.576	83.246	68.939
0.98	-36.670	32.758	83.732	71.270
0.97	-31.242	38.709	84.167	73.527
0.96	-25.857	44.422	84.549	75.708
0.95	-20.520	49.887	84.874	77.808
0.94	-15.236	55.095	85.139	79.827
0.93	-10.010	60.039	85.341	81.759
0.92	-4.848	64.709	85.476	83.603
0.91	0.246	69.096	85.541	85.355
0.90	5.264	73.191	85.532	87.012
0.89	10.202	76.984	85.445	88.570
0.88	15.053	80.468	85.275	90.025
0.87	19.810	83.633	85.018	91.376
0.86	24.467	86.469	84.668	92.617
0.85	29.017	88.968	84.222	93.745

0.84	33.452	91.120	83.673	94.758
0.83	37.765	92.918	83.016	95.649
0.82	41.948	94.352	82.245	96.417
0.81	45.993	95.413	81.353	97.057
0.80	49.892	96.095	80.335	97.565
0.79	53.635	96.387	79.183	97.937
0.78	57.214	96.284	77.889	98.169
0.77	60.620	95.777	76.446	98.256
0.76	63.843	94.859	74.847	98.196
0.75	66.874	93.525	73.081	97.982
0.74	69.702	91.768	71.140	97.612
0.73	72.317	89.583	69.015	97.081
0.72	74.709	86.966	66.695	96.385
0.71	76.867	83.912	64.171	95.519
0.70	78.780	80.419	61.430	94.479
0.69	80.437	76.485	58.462	93.261
0.68	81.827	72.109	55.254	91.861
0.67	82.938	67.291	51.793	90.274
0.66	83.758	62.032	48.067	88.497
0.65	84.277	56.335	44.062	86.526
0.64	84.482	50.204	39.763	84.357
0.63	84.362	43.646	35.155	81.986
0.62	83.905	36.668	30.224	79.411
0.61	83.100	29.279	24.953	76.628
0.60	81.936	21.490	19.327	73.634
0.59	80.402	13.315	13.329	70.427
0.58	78.488	4.769	6.942	67.005
0.57	76.183	-4.129	0.150	63.367
0.56	73.478	-13.360	-7.063	59.512
0.55	70.364	-22.900	-14.715	55.440
0.54	66.833	-32.724	-22.821	51.150
0.53	62.879	-42.805	-31.396	46.646
0.52	58.495	-53.111	-40.456	41.928
0.51	53.677	-63.609	-50.011	37.001
0.50	48.422	-74.262	-60.075	31.869

From the best fit curve of the specified n values, Table 6 shows the mean absolute error and the mean percentage error as well as the R^2 value.

Table 6. Analysis of Bike Arm Data

Participant	n	MAE (lbf*in)	MAPE (%)	MPE (%)	R^2 (%)
1	0.64	11.14	30.2	11.14	84.482
2	0.79	6.89	18.22	6.69	96.387
4	0.91	21.74	11.74	21.74	85.541
5	0.77	5.81	52.26	5.81	98.256

From Table 5, it can be calculated that for the data set the overall mean value of n is 0.78. The data set contains some errors, with participant four containing significantly more than the other three. From Figures 18, the curve of the data should be exponential as a straight line would not accurately reflect the data. This conclusion is also supported by the data collected.

For all Participants the value of MPE as shown in Table 6, is positive suggesting that the curve fit underpredicts the true values. The R^2 values show that the curve fit closely matches the data collected from Participants Four Two and Five; however, there is not enough data to make a conclusive statement that the value of n should lie around 0.77. Participant One contains the most data, and results in a R^2 value of 84.482% meaning that the curve fit generally matches the data.

The main conclusion that is drawn from the analysis of this data set, is that the curve of the line of best fit is exponential as well as the value of n is at least 0.1 higher than the average n value of the muscle characteristics used to grip calculated by the hand dynamometer if not more (which is most likely the case.

5.5 General Discussion

For the Bike dynamometer the main source of error that was found is that the maximum speed is not only a result of the maximum torque a participant output, but also their skill in pedaling a bike quickly. The other main source of error is that at no resistance tests, the bike will vibrate and move due to this vibration causing some error in the torque measurement. Overall, the bike dynamometer should have a greater exponential value of n for the theoretical equation by at least 0.1 than compared to the n value of the hand dynamometer. From the data collected by the hand dynamometer, it can be concluded that the value of n should be about 0.54. From the test data of both dynamometers, it can be concluded that the curve fit is exponential and matches the concave shape.

One of the main sources of error found in the data for the bike dynamometer is that the bike has a significant vibration at low resistances. The other main factor of error found in both dynamometers is that fatigue over the course of several tests will lower the participants maximum force output. While a more minute source of error, adrenaline can also account for some inaccuracies where the data lies significantly above the theoretical curve.

6 CONCLUSIONS

The hand dynamometer experiment successfully demonstrated that human muscle characteristics can be measured with the combination of mechanical components, sensors, data acquisition, and data analysis. By converting input force into pressure and measuring the displacement inside the cylinder in the form of electrical signals, the device allowed us to determine the force–velocity relationship of human grip strength. The use of a stiffer lock and multiple trials per person helped us to collect reliable data. Despite these improvements, a few other sources of error were present, including flinching, inconsistent grip strength, and fatigue. In addition to those, data analysis choices, such as cutoff frequency for velocity and force, influenced the outcome. This test data leads to the affirmation that n should be equal to 0.54 in the theoretical equation. The bike dynamometer experiment showed the muscle characteristics can be measured much in the same way as the hand dynamometer, but with a load cell to measure force and an IR sensor to measure velocity. From the data collected for the bike dynamometer for the leg and am inputs, the curve fit should be exponential with the value of n to be in the range of 0.64-0.8. Further testing for more data points is needed to make a more conclusive statement as to the value of n for the leg and arm characteristics.

7 FUTURE WORK

7.1 Bike Dynamometer

The accuracy of the data collected can be further improved as the bike currently has a significant vibration at no resistance applied. The suggestion for correcting this in the future is to fix it to the ground or a heavy surface. For a more accurate and certain measurement of the value of n in the theoretical equation more test data needs to be collected.

7.2 Hand Dynamometer

The reliability of the hand dynamometer can be further improved in several ways. One important next step is to improve the lock system to minimize the initial noise that occurs at the moment of latch release. This would increase the quality of the data measured by the device. Future work should also focus on data processing, primarily the selection of filtering parameters. Since cutoff frequency and sampling frequency affect the value of T/τ, and thus the degree of smoothing, a systematic way to choose them would be beneficial. In addition, averaging the force and velocity values over the region where they remain relatively constant provides a better representation of each trial than using a single data point.

Increasing the sample size and diversity of test subjects is another important step to better represent the range of human muscle characteristics. This would help to make meaningful comparisons among different individuals.

8 ACKNOWLEDGEMENTS & REFERENCES

Dr. Robert Woods – Faculty Advisor

John Skiba – Graduate Student, former member of the Hand Dynamometer group

References

[1] Witkowski, Grayson, 2024, “Hand Dynamometer,” The University of Texas at Arlington, Arlington, TX

9 NOMENCLATURE

Greek Symbols

θ	Angle or angular position
θ_{pulse}	Angular distance between consecutive detected edges
θ_{strip}	Angular width of one vinyl strip
τ	Torque; also used as the first-order filter time constant
τ_a	Applied torque
τ_{out}	Output torque
$\tau_{flywheel}$	Braking torque acting on the flywheel
ω	Angular velocity
ω_c	Cutoff angular frequency
π	Mathematical constant pi

Roman Symbols

A	Cross-sectional area
A_{piston}	Cross-sectional area of the piston
C	Flywheel circumference
Count	Number of detected IR-sensor edges
D	Diameter
$D_{cylinder}$	Inside diameter of the hand-dynamometer cylinder
$D_{piston\ rod}$	Diameter of the piston rod
d	Flywheel diameter
e	Base of the natural logarithm
F	Force
Fa	Force applied to the pedal
$F_{applied}$	Force applied by the participant
F_f	Friction force between the brake pad and flywheel
F_{load}	Force measured by the load cell
F_{max}	Maximum force
$F_{measured}$	Experimentally measured force
$F_{theoretical}$	Force predicted by the theoretical model
f	Sampling frequency
fc	Filter cutoff frequency
G(s)	Transfer function
$G_{driving}$	Number of teeth on the driving sprocket
G_{driven}	Number of teeth on the driven sprocket
i	Data-point index
k	Discrete sample index
L	Length or moment-arm distance

L1-L7	System-specific lengths and moment arms identified
L_{stroke}	Total piston stroke length
$L_{displacement}$	Piston displacement used in the calculation
M	Moment about a specified point
n	Exponent controlling the shape of the theoretical force-velocity
N	Total number of data points
$N_{driving}$	Number of teeth on the driving sprocket
N_{driven}	Number of teeth on the driven sprocket
P	Pressure inside the hand-dynamometer cylinder
P_{avg}	Average initial pressure-voltage reading
P_{mult}	Pressure-to-force conversion multiplier
R	Reaction force
R_{load}	Reaction force measured by the load cell
R^2	Coefficient of determination
r	Radius of the flywheel
s	Length of one vinyl strip; also the Laplace-domain variable when used in G(s)
t	Time
T	Sampling period
T_{stroke}	Time required to complete the piston stroke
V	Velocity
V_k	Filtered velocity at the current sample
V_{max}	Maximum velocity
V_p	Measured pressure-transducer voltage
$V_{raw,k}$	Raw velocity at the current sample
V_{load}	Measured load-cell voltage
V_x	Measured position-sensor voltage
$V_{displacement}$	Displaced fluid volume
W	Weight of the brake assembly
X	Position or displacement
X_{avg}	Average initial position-voltage reading
X_{mult}	Position-voltage conversion multiplier
x(t)	Input signal to the filter
xk	Input signal at the current sample
y(t)	Output signal from the filter
yk	Filtered output at the current sample

Subscripts

a	Applied
applied	Applied by the participant
avg	Average value
c	Cutoff
cylinder	Hand-dynamometer cylinder

driven	Driven sprocket
driving	Driving sprocket
f	Friction
i	ith data point
k	Current discrete sample
k-1	Previous discrete sample
k+1	Next discrete sample
load	Load-cell measurement
max	Maximum value
measured	Experimentally measured value
n	Current detected edge or measurement
n-1	Previous detected edge or measurement
out	Output
p	Pressure measurement
piston	Piston
pulse	Distance between consecutive detected edges
raw	Unfiltered value
strip	Vinyl strip
theoretical	Theoretical or predicted value
x	Position measurement

Acronyms and Abbreviations

CAD	Computer-Aided Design
DAQ	Data Acquisition
FBD	Free-Body Diagram
IR	Infrared
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MPE	Mean Percentage Error
NI	National Instruments
RPS	Revolutions per Second
RSE	Referenced Single-Ended
VI	Virtual Instrument

Units

Hz	Hertz
in	Inch
in^2	Square inch
in^3	Cubic inch
lbf	Pound-force
lbf·in	Pound-force inch
mm	Millimeter

mm^2	Square millimeter
mm^3	Cubic millimeter
mm/s	Millimeters per second
psi	Pounds per square inch
rad	Radian
rev/s	Revolutions per second
s	Second
V	Volt

10 APPENDIX A: DATA PROCESSING

For all test data collected two sheets are created in Excel file, one containing the raw data and the other containing a cut version of the raw data. The cut version should include only the data of the run of interest for analysis. The cut data should start at constant pressure voltage for at least five data points before the pressure increases significantly signaling the start of a test. An example of how the data should be cut can be found in Table 3.

Table 7. Top Cut of Data

Position (V)	Pressure (V)
4.101025	1.007579
4.101025	1.008546
4.101348	1.008224
4.101025	1.009514
4.101025	1.009514
4.100703	1.009836
4.100703	1.010158
4.100381	1.010481
4.100381	1.01177
4.099736	1.01306
4.096835	1.024342
4.071368	1.188106
4.047835	2.203888
4.014308	3.730624

More data points can be taken away from the initial increase of pressure, but at least five need to be taken at a steady pressure reading before the test begins. For the bottom cut of the raw data, only the data point where the test has ended needs to be recorded. This will be determined when the potentiometer reads a voltage between 1.9 - 2.0. More points after the test can be saved but there is no need as those specific data points hold no relevance other than signifying the end of a test. An example of the bottom cut of the data can be found in Table 4.

Table 8. Bottom Cut of Data

Position (V)	Pressure (V)
2.077197	1.178112
2.066237	1.167474
2.056243	1.156836
2.04625	1.145875
2.036901	1.131691

2.027875	1.114606
2.019816	1.096875
2.011112	1.080112
2.00273	1.063027
1.995638	1.049487
1.987901	1.034981
1.980809	1.022731
1.974039	1.010481
1.966947	1.003066

The cut data should then be copied into the zero-phase filter sheet as shown in Figure 14.

	A	B	C	D	E	F	G	H	I	J	K	L
1	T/tau =	0.95		T/tau =	0.95							
2	exp =	0.3867		exp =	0.3867							
3	1-exp =	0.6133		1-exp =	0.6133							
4												
5	Insert Data:				Backward			Forward			Filtered	
6		X (volt)	P (volt)									
7		4.1133	1.0321		4.1133	1.0321		4.1127	1.0328		4.113	1.0324
8		4.1123	1.0305		4.1127	1.0311		4.1119	1.0338		4.1123	1.0325
9		4.1126	1.0321		4.1127	1.0317		4.1113	1.0392		4.112	1.0354
10		4.112	1.0324		4.1122	1.0321		4.1091	1.0505		4.1107	1.0413
11		4.1101	1.0359		4.1109	1.0345		4.1044	1.0792		4.1077	1.0568
12		4.1049	1.0485		4.1072	1.0431		4.0955	1.1477		4.1014	1.0954
13		4.0888	1.1185		4.0959	1.0893		4.0807	1.3051		4.0883	1.1972
14		4.0759	1.3267		4.0836	1.2349		4.0679	1.6009		4.0758	1.4179
15		4.062	1.7726		4.0704	1.5646		4.0552	2.0357		4.0628	1.8002
16		4.0491	2.5537		4.0573	2.1712		4.0444	2.4531		4.0509	2.3121
17		4.0433	2.6813		4.0487	2.484		4.0369	2.2936		4.0428	2.3888
18		4.032	1.8967		4.0385	2.1238		4.0266	1.6788		4.0325	1.9013
19		4.0249	1.4189		4.0302	1.6915		4.018	1.3333		4.0241	1.5124
20		4.014	1.2207		4.0203	1.4028		4.0069	1.1975		4.0136	1.3001
21		4.0037	1.1559		4.0101	1.2514		3.9956	1.1608		4.0028	1.2061
22		3.9927	1.1475		3.9994	1.1877		3.9827	1.1687		3.9911	1.1782
23		3.9776	1.1697		3.986	1.1767		3.9669	1.2023		3.9765	1.1895
24		3.9602	1.2149		3.9702	1.2001		3.9501	1.2539		3.9601	1.227
25		3.9444	1.2855		3.9543	1.2524		3.9341	1.3159		3.9442	1.2842
26		3.9273	1.3532		3.9377	1.3142		3.9178	1.3642		3.9278	1.3392
27		3.9131	1.3857		3.9226	1.3581		3.9028	1.3818		3.9127	1.3699
28		3.8966	1.3851		3.9067	1.3746		3.8865	1.3755		3.8966	1.3751
29		3.8802	1.3644		3.8904	1.3684		3.8704	1.3603		3.8804	1.3643
30		3.8654	1.347		3.8751	1.3553		3.8548	1.3538		3.8649	1.3545
31		3.8483	1.3519		3.8586	1.3532		3.838	1.3644		3.8483	1.3588
32		3.8318	1.377		3.8422	1.3678		3.8216	1.3843		3.8319	1.3761
33		3.8164	1.4009		3.8264	1.3881		3.8055	1.3959		3.8159	1.392
34		3.7993	1.408		3.8098	1.4003		3.7882	1.3881		3.799	1.3942
35		3.7809	1.388		3.7921	1.3927		3.7705	1.3565		3.7813	1.3746
36		3.7642	1.3428		3.7749	1.3621		3.754	1.3066		3.7645	1.3344
37		3.748	1.2829		3.7584	1.3135		3.7378	1.2491		3.7481	1.2813
38		3.7319	1.2197		3.7422	1.256		3.7215	1.1956		3.7319	1.2258
39		3.7155	1.1672		3.7258	1.2015		3.7051	1.1573		3.7154	1.1794
40		3.6984	1.1336		3.709	1.1599		3.6886	1.1417		3.6988	1.1508
41		3.6832	1.1278		3.6932	1.1402		3.6731	1.1544		3.6831	1.1473
42		3.6674	1.153		3.6774	1.148		3.657	1.1965		3.6672	1.1723
43		3.6513	1.211		3.6614	1.1866		3.6405	1.2655		3.6509	1.2261
44		3.6342	1.3019		3.6447	1.2573		3.6233	1.352		3.634	1.3047

Auto-GraphH | Auto-Filter | cut | raw

Figure 22. Excel Zero Phase Filter

Then convert the filtered data into the position and force measurements as shown in Appendix C. The test data will generally converge into three types of shapes, one for low resistance tests and two for high resistance tests. For low resistance tests, the data point to be taken should occur after the dynamic effect, but before the velocity drops off. Note that for both high and low resistance test data the data point should be taken at a generally constant velocity. The considerations for where to take the data point for low resistance test data can be seen in Figure 23.

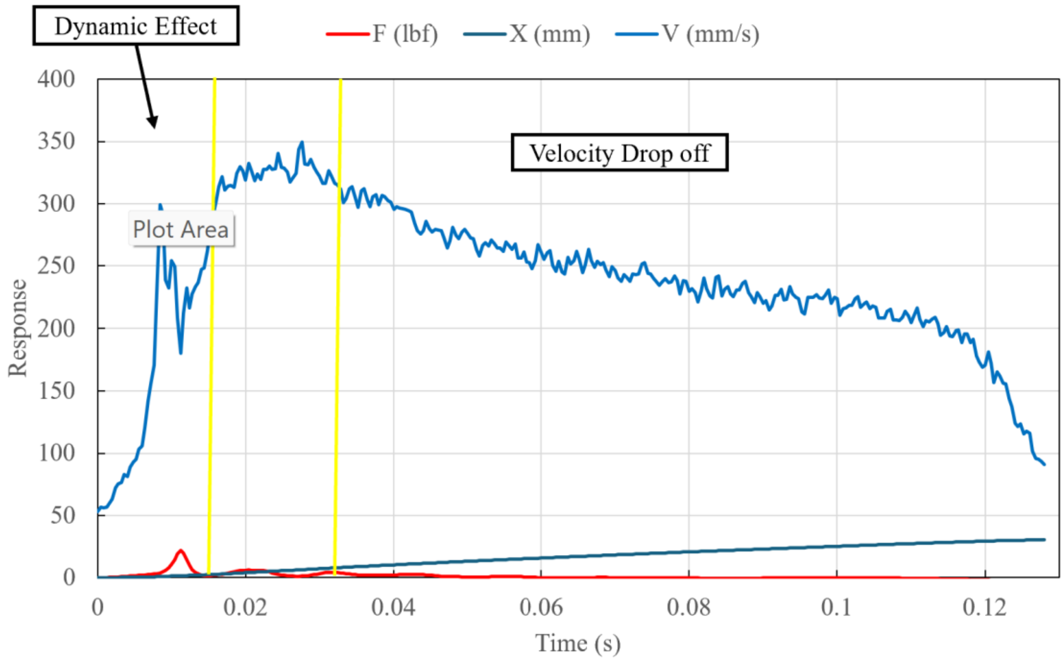


Figure 23. Low Resistance Example Data Graph

The shape of a high resistance run will have a flat velocity and force curve after the dynamic effect or one with a single hump, and the velocity decreasing at the end like in the low resistance tests. The data point for graphs of these characteristics should be anywhere at constant velocity or at the peak of the hump but before velocity drops off as seen in Figures 24-25.

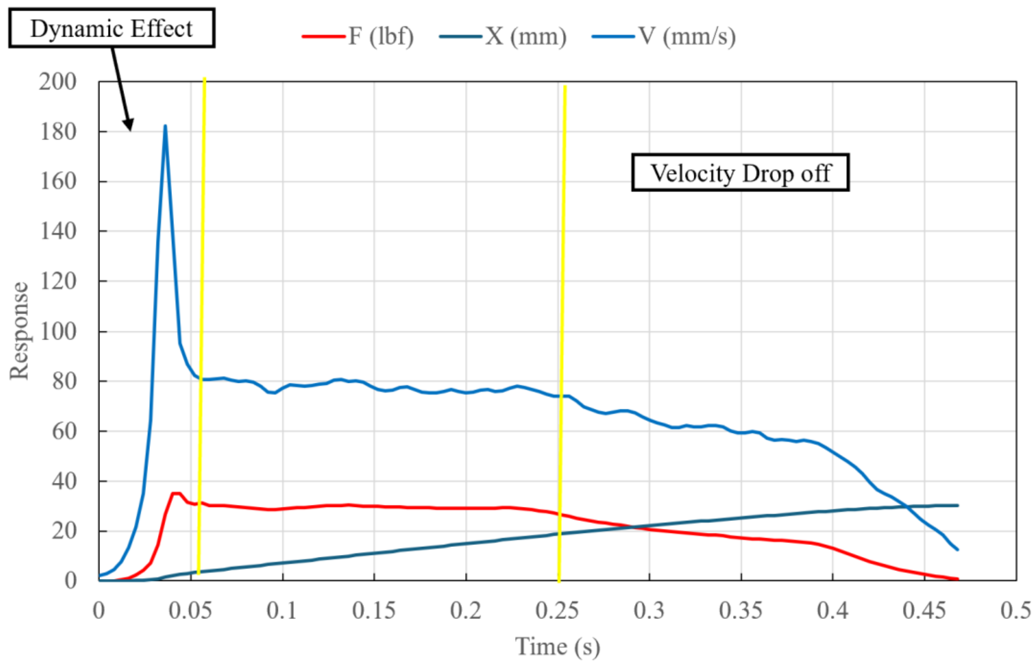


Figure 24. High Resistance Example Data Graph 1

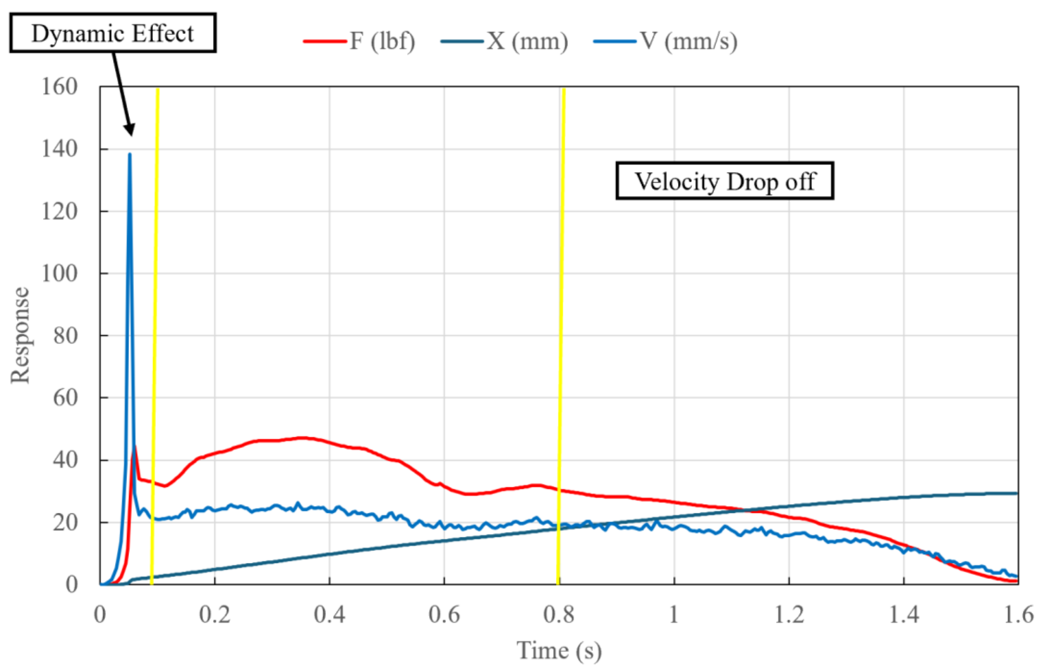


Figure 25. High Resistance Example Data Graph 2

Each individual data point from every test should then be placed in a Force vs. Velocity table to be graphed. Along with the data point from a collective of tests, the Force vs. Velocity graph as seen in Figure 20 also contains the theoretical curve found by Equation 2 where n is equal to 0.54:

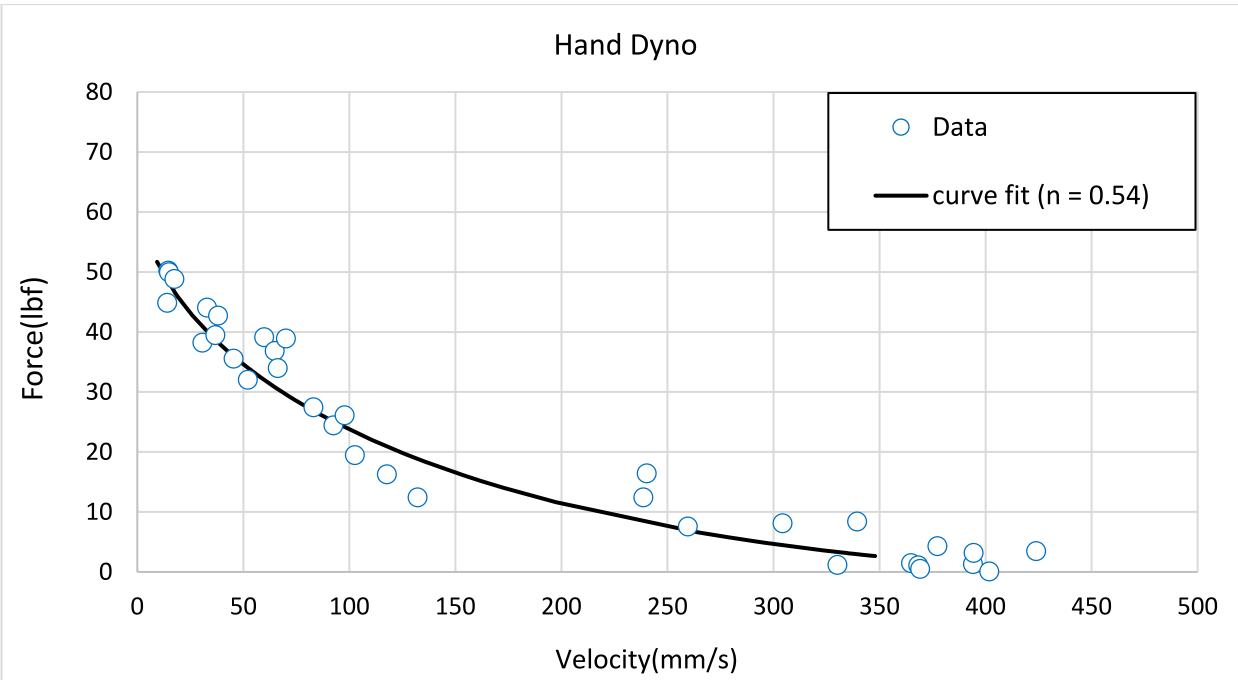


Figure 26. Force vs. Velocity Graph Example

The values used for the maximum force should be the recorded maximum force of the participant. The maximum velocity used in the formula should typically be around 500 mm/s but can be lowered to around 450 mm/s depending on the participant.

11 APPENDIX B: CALIBRATION

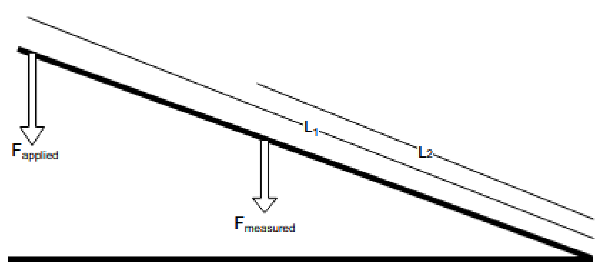
CALIBRATION EQUATIONS FOR HAND DYNAMOMETER

Displacement Calibration

$$X(\text{mm}) = 18.4\,(4.10 - V_x)$$

Where:
X(mm)= displacement in millimeters
V_x= measured position voltage (V)
4.10 V = zero-displacement reference voltage
18.4 mm/V = displacement calibration factor

Force Calibration



Knowns:

- $A = 1.62\ \text{in}^2$ (area of piston)
- $L_2 = 17\ \text{in}$
- $L_1 = 24.5\ \text{in}$

$$F_{\text{applied}} = \frac{L_1}{L_2} \cdot F_{\text{measured}}$$

$$F_{\text{measured}} = P \cdot A$$

$$F_{\text{applied}} = \frac{L_1}{L_2} \cdot (P \cdot A)$$

$$\frac{L_1}{L_2} = \frac{24.5}{17} = 1.441$$
$$F_{\text{applied}} = 1.441 \cdot (P \cdot 1.62)$$
$$F_{\text{applied}} = 2.34 \cdot P$$

$$\boxed{F_{\text{applied}} = 2.34\,P}$$

where:

- P is in **psi**
- F_{applied} is in **lbf**

Including Voltage Calibration

$$P = 25(V_p - 1)$$
$$F_{\text{applied}} = 2.34 \cdot 25(V_p - 1)$$

$$\boxed{F_{\text{applied}} = \mathbf{58.5(\boldsymbol{V_p} - 1)}}$$

Where:

$F_{applied}$ = applied force by the user (lbf)
$F_{measured}$ = force generated by fluid pressure (lbf)
P = pressure inside the cylinder (psi)
A = piston cross-sectional area (in^2)
L_1 = distance from pivot to applied force location (in)
L_2 = distance from pivot to measured force location (in)
V_p = measured pressure voltage (V)
1 V = zero-pressure reference voltage

25 psi/V = pressure calibration factor

First-Order Low Pass Filter

Continuous Form:

$$\tau \frac{dy}{dt} + y(t) = x(t)$$

Transfer Function:

$$G(s) = \frac{1}{\tau s + 1}$$

Cutoff Frequency:

$$\omega_c = \frac{1}{\tau}$$

$$f_c = \frac{1}{2\pi\tau}$$

Sampling Frequency:

$$f \text{ (Hz)}, T = \frac{1}{f}$$

Discrete Implementation:

$$y_k = y_{k-1} + \left(1 - e^{-T/\tau}\right)(x_k - y_{k-1})$$

Backward Difference

$$f(x_k - x_{k-1})$$

Forward Difference

$$f(x_{k+1} - x_k)$$

Central Difference

$$\frac{f}{2}(x_{k+1} - x_{k-1})$$

Filtered Velocity Equation

$$V_k = V_{k-1} + \left(1 - e^{-\frac{1}{f\tau}}\right)\left(V_{\mathrm{raw},k} - V_{k-1}\right)$$

HAND DYNAMOMETER CYLINDER:

Parameters

$$D_{\mathrm{cylinder}} = 45\ \mathrm{mm}$$
$$D_{\mathrm{piston\ rod}} = 16\ \mathrm{mm}$$
$$L_{\mathrm{stroke}} = 40\ \mathrm{mm}$$
$$L_{\mathrm{displacement}} = 20\ \mathrm{mm}$$
$$T_{\mathrm{stroke}} = 0.15\ \mathrm{s}(\text{ideal time for zero resistance})$$

Piston Area

$$A_{\mathrm{piston}} = \frac{\pi}{4}\left(D_{\mathrm{cylinder}}^2 - D_{\mathrm{piston\ rod}}^2\right)$$
$$A_{\mathrm{piston}} = \frac{\pi}{4}(45^2 - 16^2)$$
$$A_{\mathrm{piston}} = 1389.4\ \mathrm{mm}^2 = 2.15\ \mathrm{in}^2$$

Displacement Volume

$$V_{\mathrm{displacement}} = A_{\mathrm{piston}} \times L_{\mathrm{displacement}}$$
$$V_{\mathrm{displacement}} = 27{,}787.4\ \mathrm{mm}^3 = 1.70\ \mathrm{in}^3$$

12 APPENDIX C: EXCEL DOCUMENT SETUP

Each Excel document contains four sheets to analyze individual tests: Raw Data, Cut Data, Auto-Filter, and Auto Graph. The Raw Data sheet is the data exported by LabVIEW in three columns: Time, Voltage_0 (Position), and Voltage_1 (Pressure). The Cut Data sheet is just a page to contain the trimmed version of the raw data to preserve all of the original data, but also to remove the data points that serve no purpose. An example of the Raw Data sheet can be seen in the figure below.

Time	Voltage_0	Voltage_1
00.000	4.133262	1.006612
00.001	4.132295	1.00629
00.002	4.132617	1.006612
00.003	4.132617	1.006612
00.004	4.133262	1.006612
00.005	4.133262	1.006612
00.006	4.133262	1.005967
00.007	4.13294	1.006612
00.008	4.13294	1.005645
00.009	4.133262	1.00629
00.010	4.13294	1.005967
00.011	4.133262	1.005967
00.012	4.133262	1.005645
00.013	4.13294	1.00629
00.014	4.132617	1.005645
00.015	4.133262	1.006935
00.016	4.132617	1.00629
00.017	4.13294	1.006612
00.018	4.13294	1.005967
00.019	4.133262	1.00629
00.020	4.133262	1.005967
00.021	4.133262	1.006612
00.022	4.133262	1.00629
00.023	4.132617	1.005967
00.024	4.13294	1.00629
00.025	4.13294	1.005967
00.026	4.133262	1.00629
00.027	4.133262	1.006612
00.028	4.133262	1.00629
00.029	4.13294	1.006935
00.030	4.133585	1.00629
00.031	4.133262	1.00629
00.032	4.13294	1.00629
00.033	4.132617	1.006935
00.034	4.133585	1.00629
00.035	4.133262	1.006935

Figure 27. Example of Raw Data Sheet in Excel

The Auto-Filter sheet is where the cut data is then filtered by the zero-phase filter as shown in Appendix A. For the zero-phase filter, the user inputs the two exponential value labeled T/tau that is calculated below as 1-exp(T/tau). There is an exponential value for the position and pressure voltage data. For the Backward section of the zero phase filter the first value is defined by the formula =IF(ISNUMBER(B7),B7,""). This formula detects if there is a value in the inserted data column and then assigns the cell the same value as the number in the inserted data column. All values after the first in the Backward column have the formula =IF(ISNUMBER(B8),E7*B2+B8*B3,"") with increasing row values. This formula detects if there is a value in the Insert Data column in the same row, and if so, it calculates:

$$f = x_{n-1}\left(e^{T/\tau}\right) + x_n\left(1 - e^{T/\tau}\right)$$

The Forward Column is calculated by the formula =IF(AND(ISNUMBER(B7), ISBLANK(B8)), B7, IF(AND(ISNUMBER(B8), ISNUMBER(B7)), B2*H8+B3*B7, "")), where the row values increase. This formula calculates where the last row in the Insert Data column contains a value and then assigns that value to the cell in the Forward column. If the row isn't the last row, it calculates the formula:

$$f = x_{n+1}\left(e^{T/\tau}\right) + x_n\left(1 - e^{T/\tau}\right)$$

The Auto-Graph Sheet is shown in Figure # below.

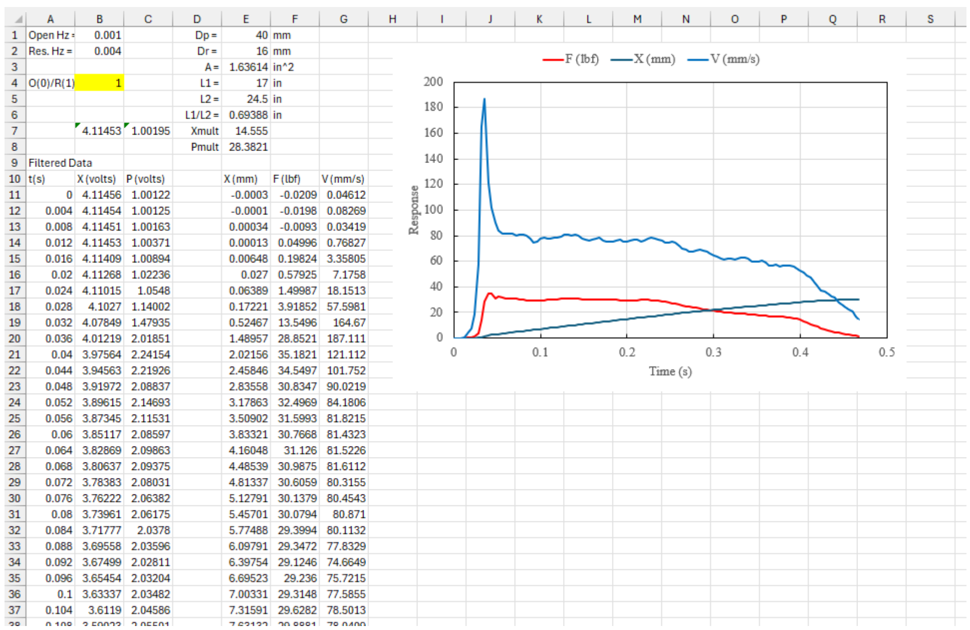


Figure 28. Auto-Graph Sheet

Like in the Auto-Filter Sheet, the X(mm), F(lbf), and V(mm/s) are set to have an if condition to only initiate the cells formulas if there is a corresponding data point in the X(volts) and P(volts) on the same row. To convert the voltages into mm and lbf, some multiples were created at the top of the page. Xmult contains the equation =2*25.4/5.03/(L1/L2), and Pmult contains the equation =(100/4)*A*(L1/L2). The reasons Xmult and Pmult are assigned to these equations can be found in the math of Appendix B. The X(mm) cells are calculated by the formula (Xavg-X(volts))*Xmult, where Xavg is the average of the first five X(volts) values. The P(lbf) cells are calculated by the formula (P(volts)-Pavg)*Pmult, where Pavg is the average of the first five P(volts) values. The Velocity column uses the equation $V(n) = \frac{X_{n+1} - X_{n-1}}{2(\frac{1}{frecuency})}$. At the first and last rows of the velocity calculation the formula only uses the current position minus the previous or next position divided by the change in time.

13 APPENDIX D: BIKE DYNAMOMETER LABVIEW PROGRAM

The LabVIEW program was written to provide the data required to make a torque versus velocity curve. The most basic requirement of the program was that it recorded the time at which an edge was detected and the force recorded by the load cell. Another requirement for the angular velocity calculation was that the calculated angular velocity should be instantaneous. While an average angular velocity could be used, the team preferred to collect the instantaneous angular velocity for analysis. Calculating the revolutions per second and the torque could be performed in Excel since the data would already be output to an Excel file. By performing further calculations in Excel, the LabVIEW program remained simpler and parameters such as the load cell offset could be modified more easily. Based on these requirements, two primary concepts were considered when designing the program:

1. Record three columns of data: time, force, edge detection. This would output the force and a zero or one for edge detection at every time a sample was taken by the DAQ.
2. Record two columns of data: time of edge detection, force. This would output the time and force whenever the infrared sensor detected an edge.

Of these two approaches, the second was selected as the more effective solution because it avoided several inherent limitations of the first approach. The primary drawback of the first approach was that it required a significantly larger volume of data to be stored to achieve the same result (resulting in more rows in Excel and a larger array in LabVIEW). Additionally, the time intervals between edge detections would contain zeros. These rows of zeros would significantly complicate the formulas needed to calculate revolutions per second in Excel, as an additional column containing only the non-zero angular velocity values would be required, resulting in a different row count from the torque data.

As a result, the final LabVIEW program followed the logic of the second concept, with an extra addition that it would display the highest force detected from the load cell to measure the stall torque. The program can be seen in Figures 30-31. The following paragraphs describe the inner workings of the program to facilitate any future modifications that would be desired at a later date.

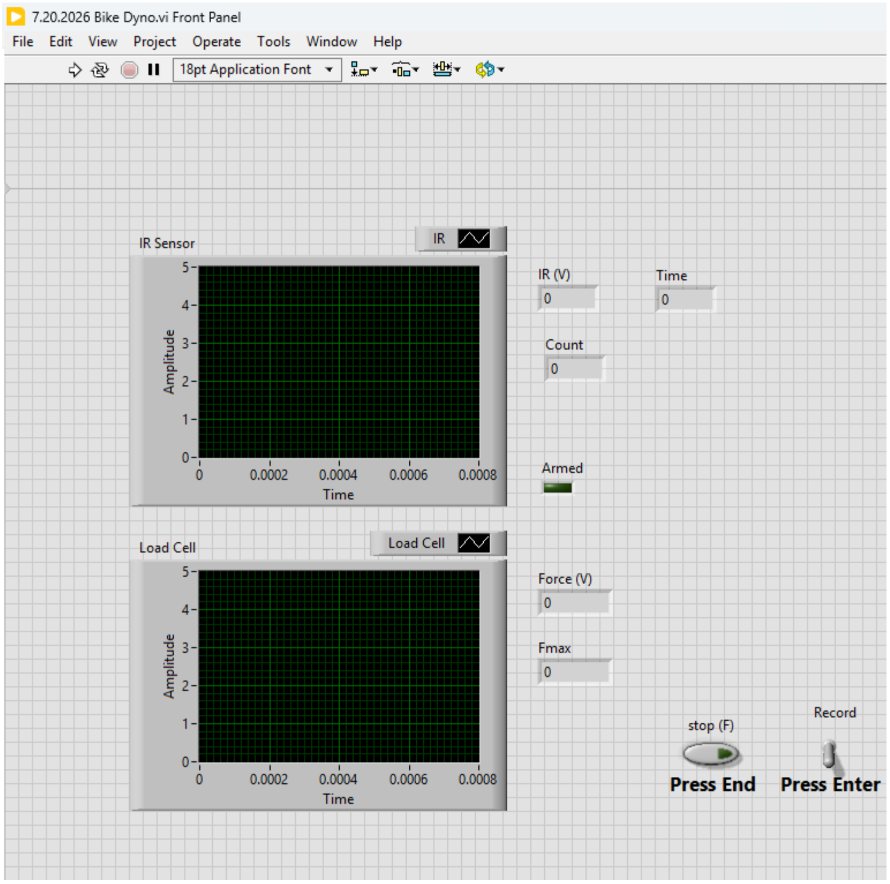


Figure 29. LabVIEW Program Front Pannel

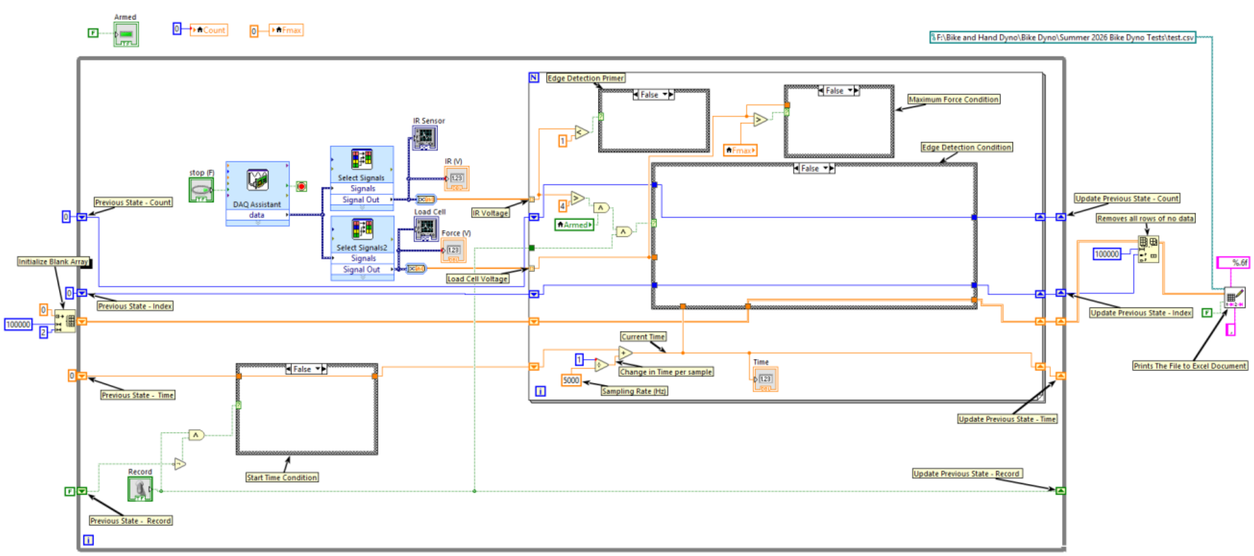


Figure 30. LabVIEW Program Block Diagram false conditions

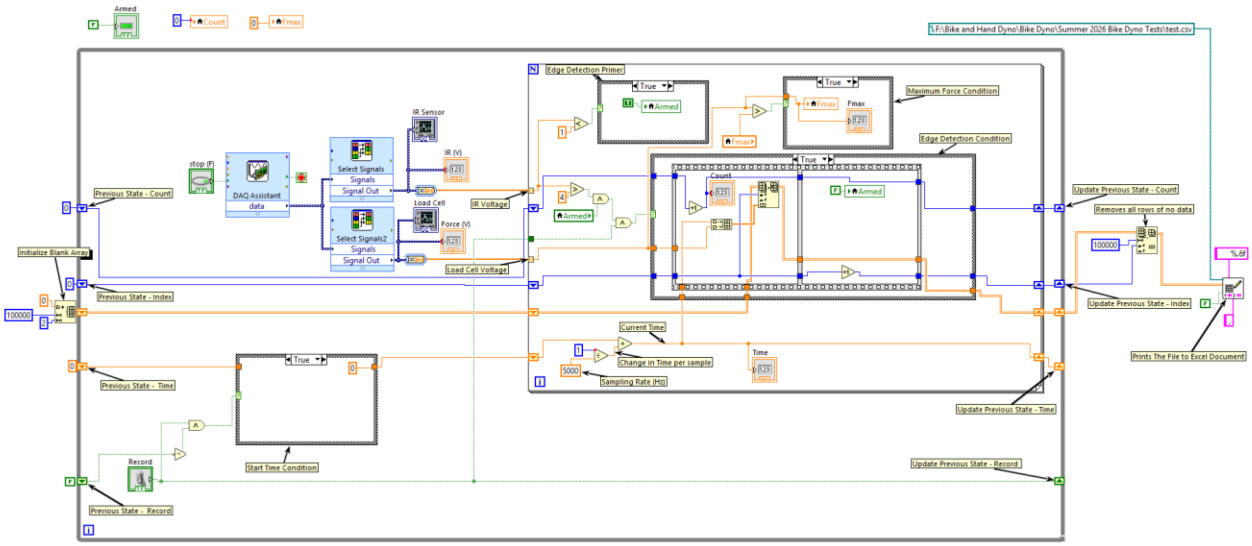


Figure 31. LabVIEW Program Block Diagram true conditions

The DAQ Assistant is configured in data collection mode with continuous sampling, which is controlled by two parameters: Samples to Read and Sample Rate. The Sample Rate parameter determines the number of samples collected per second (Hz), while Samples to Read determines the number of samples output as an array of scalar values. It is important to note that although the DAQ Assistant is located within the while loop, its data acquisition timing is independent of the while loop execution rate. The DAQ can be considered a hardware-level component with its own internal timing system; therefore, it collects data samples at the rate specified by the Sample Rate parameter and outputs an array of scalar values only after the number of collected samples equals the value specified by Samples to Read. For this specific program the Samples to Read is set to 5 and Sample Rate is set to 5,000. As shown in Figure 32, the IR sensor and load cell are in Terminal Configuration RSE, which is the proper setting for any hardware that is amplified.

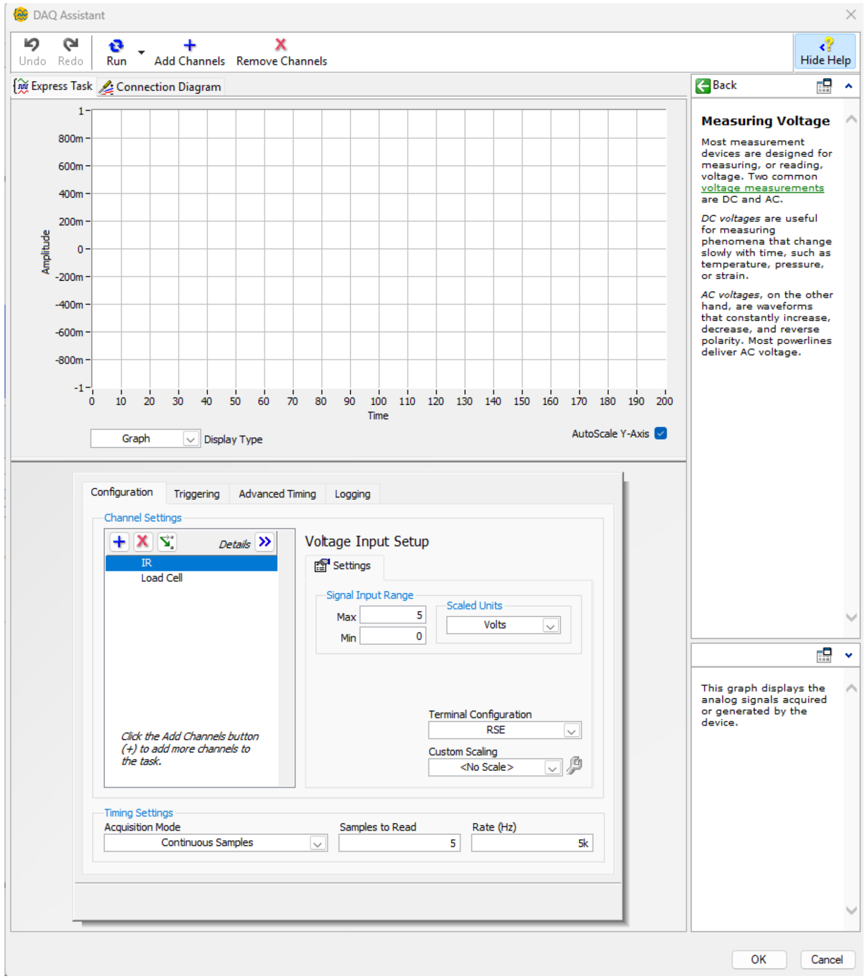


Figure 32. DAQ Assistant Configuration

After the DAQ Assistant collects the data, the Select Signals blocks separate the total data output into two individual 1D arrays of scalar values. These two arrays correspond to the recorded voltages from the IR sensor and the load cell. These separate signals are then individually analyzed within the for-loop. At the border of the for-loop, there is a small block containing []', which represents the auto-indexing feature of LabVIEW. This block extracts the scalar value from the signal array at the corresponding index for each iteration of the for-loop. Because the for-loop is configured for auto-indexing, it begins processing at the first element of the signal array and continues sequentially until the final element has been analyzed.

Before discussing the program logic in greater detail, it is important to explain the function of the shift registers used throughout the program. Shift registers are the small blocks located on the borders of the for-loop and while loop, identified by either an upward or downward arrow. The blocks containing a downward arrow are referred to as left shift registers, while the blocks containing an upward arrow are referred to as right shift registers. Shift registers are used to store previous values of signals or variables between loop iterations, allowing values to persist between executions of the loop, or they can function as variables within the program. The left side of the left shift register is used to initialize the stored value, while the left side of the right shift register updates the stored value by receiving the output from the previous iteration. The right side of both shift registers outputs the current stored value. Within this program, the variables that require storage through shift registers are the Record variable and the Blank Array. The count, index, and time variables are also implemented using shift registers; however, these variables could alternatively be replaced with local variables if desired.

With the operation of the individual LabVIEW components established, the overall program logic can now be explained. The collected data from the DAQ Assistant is passed into the for-loop, where the IR sensor and load cell data are analyzed. When the IR voltage is less than one volt, the Armed variable is assigned to a value of one, indicating that the system is ready to detect an edge. When the IR voltage exceeds four volts, and both Armed and Record are true, the Edge Detection condition is triggered. This detection method identifies the leading-edge transition, which occurs when the sensor changes from detecting a reflective surface to a non-reflective surface. The large voltage difference threshold, combined with the Armed variable, helps prevent false detections.

During testing, alternative edge detection methods were evaluated but were found to be unreliable. Specifically, limiting detection to cases where the difference between the current and previous voltage values exceeded two volts, or where the previous value was less than two volts while the current value exceeded four volts. These methods could only detect an edge during a single completion of the for-loop and failed to transfer the detection state between subsequent for-loop executions. This prevented edge detection between the final and first iterations of consecutive for-loops and resulted in false detections caused by voltage fluctuations during operation. The current implementation, shown in Figures 29-31, utilizes the Armed variable to maintain the trigger condition between for-loop executions, allowing for a larger voltage threshold and more reliable edge detection.

When an edge is detected, the Edge Detection condition shown in Figures 30-31 is executed. During this condition, the current time and load cell value are stored together using a Build Array block. The resulting data pair is then inserted into the Blank Array using a Replace Array Subset block at the specified index. Once the data has been stored, the program resets Armed to false and increases the Index value by one. Similarly, the Count variable is increased to provide a visual indication that the program is operating correctly; however, this value is not included in the recorded Excel output.

The time value used for the recorded data is generated from the hardware timer within the DAQ Assistant. Therefore, the timer begins at zero and increases by 0.0002 seconds per for-loop iteration, corresponding to the inverse of the Sample Rate (1/Sample Rate). The Start Time condition is only activated at the beginning of data collection, which occurs when the Previous Record value is false and the Record value changes to true. When triggered, this condition resets the time value to zero to ensure that each data collection cycle begins with an accurate time reference.

The stop (F) Boolean condition controls the termination of the while loop. Once the while loop terminates, the Blank Array, which now contains the recorded data values, is passed to a Delete From Array block. This block removes all unused rows after the final recorded value, reducing the output array size to contain only the necessary data. The resized array is then passed to the

Write Delimited Spreadsheet block, which exports the data into an Excel file. For this reason, the stop (F) button must be used to output the recorded data to Excel rather than the abort program button located in the LabVIEW toolbar which will delete all recorded data without sending it to Excel.

Finally, the Maximum Force display is controlled through the Fmax variable. The Fmax variable is initially set to zero and is continuously updated whenever the current recorded force value exceeds the previously stored Fmax value. This ensures that the display continuously represents the maximum force measured by the load cell during the test.

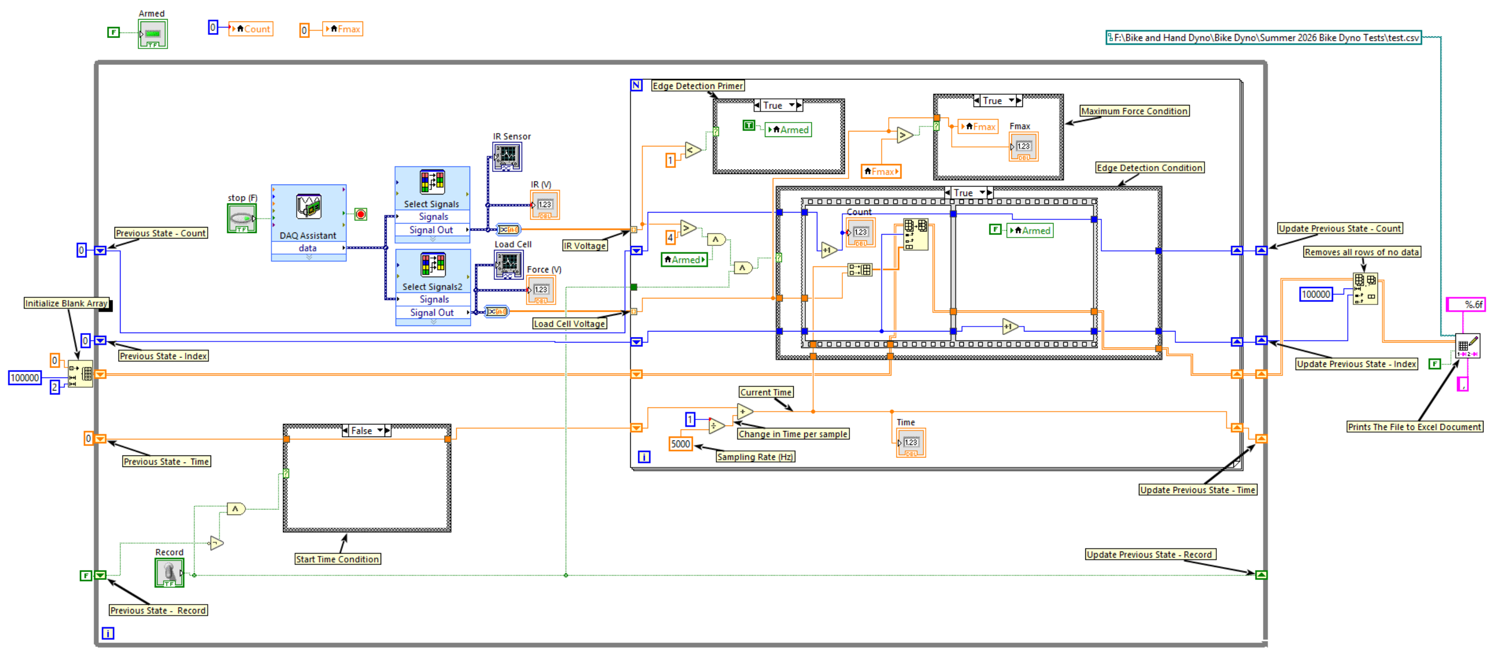


Figure 33. Large Display of LabVIEW Program Block Diagram

14 APPENDIX E: LOAD CELL SIZING

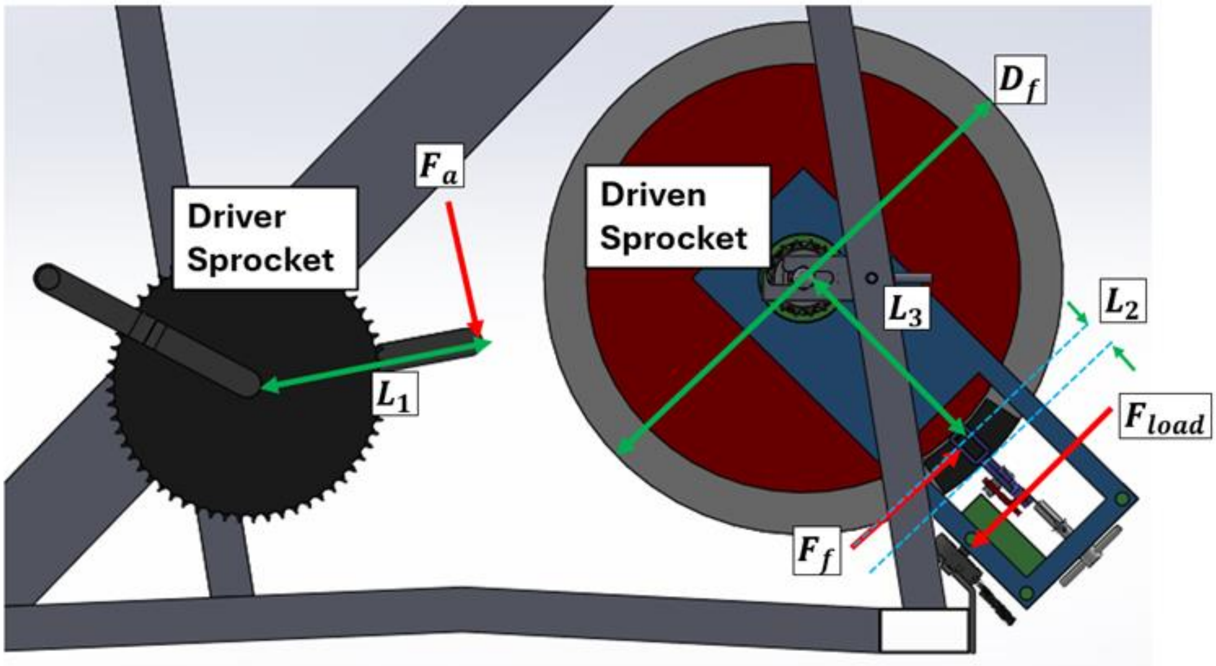


Figure 34. Forces applied on bike

Assumptions:
1) No frictional losses between chain and sprockets
2) Symmetry is assumed
3) Bike is stalled

The following parameters were measured and used for calculation:

Parameters	Description	Quantity	Units
Ndriving	Number of Gears of Driving Sprocket	52	unitless
Ndriven	Number of Gears of Driven Sprocket	16	unitless
Df	Diameter of Flywheel	18	in
L1	Length of Pedal	6 5/8	in
L2	Length from Friction Force to Edge of Flywheel	5/8	in
L3	Length of Friction Force to Center of Flywheel	8 3/8	in
Fa	Applied Force to Pedal	180	lbf

Figure 35. Parameters for loadcell sizing

Starting with the Gear Ratio:

$$Gear\ Ratio = \frac{N_{Driving}}{N_{Driven}} = \frac{\tau_a}{\tau_{out}}$$

Static equilibrium of driven sprocket and flywheel gives:

$$\tau_{out} = F_f L_3$$

The applied torque is equal to:

$$\tau_a = F_a L_1$$

Combining equations gives:

$$\frac{G_{Driving}}{G_{Driven}} = \frac{F_a L_1}{F_f L_3}$$

$$F_f L_3 \frac{G_{Driving}}{G_{Driven}} = F_a L_1$$

Solving for friction gives

$$F_f = \frac{F_a L_1 G_{Driven}}{L_3 G_{Driving}}$$

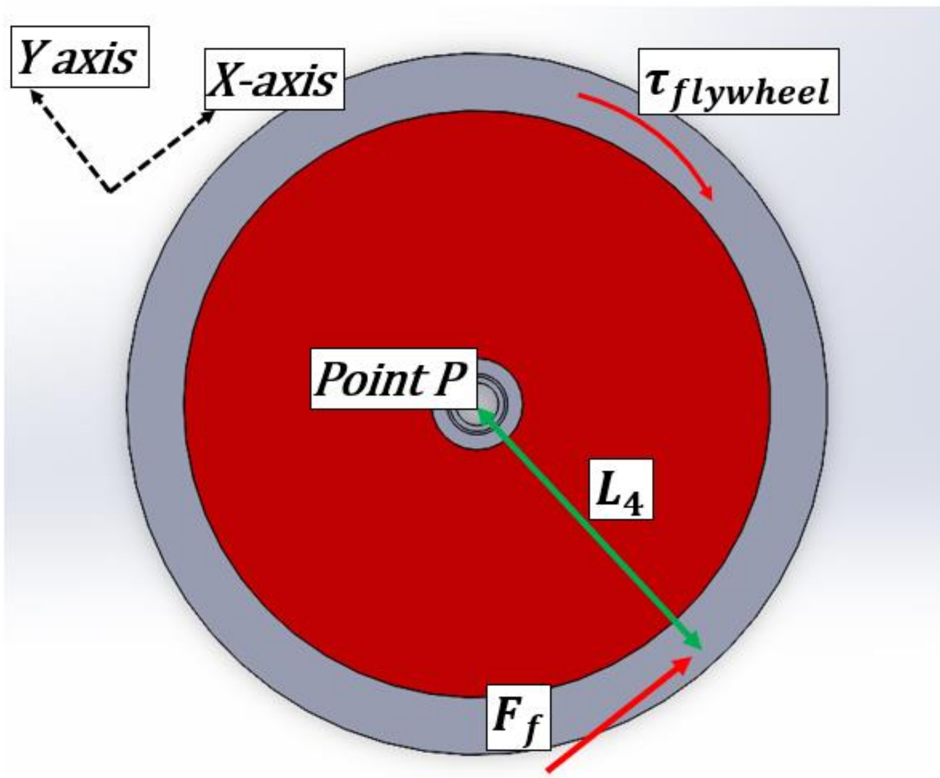
To calibrate the load cell before the flywheel rotates, the load cell will be zeroed out eliminating the weight term in the equation. The lengths were estimated based on the CAD of the bike. To estimate the max friction force to be applied, 180 applied forces to pedal were used.

$$F_f = \frac{(180lbf)(6\frac{5}{8}in)(16)}{(8\frac{3}{8}in)(52)}$$

$$F_f = 44\ lbf$$

15 APPENDIX F: LOAD CELL FORCE TO BRAKE TORQUE

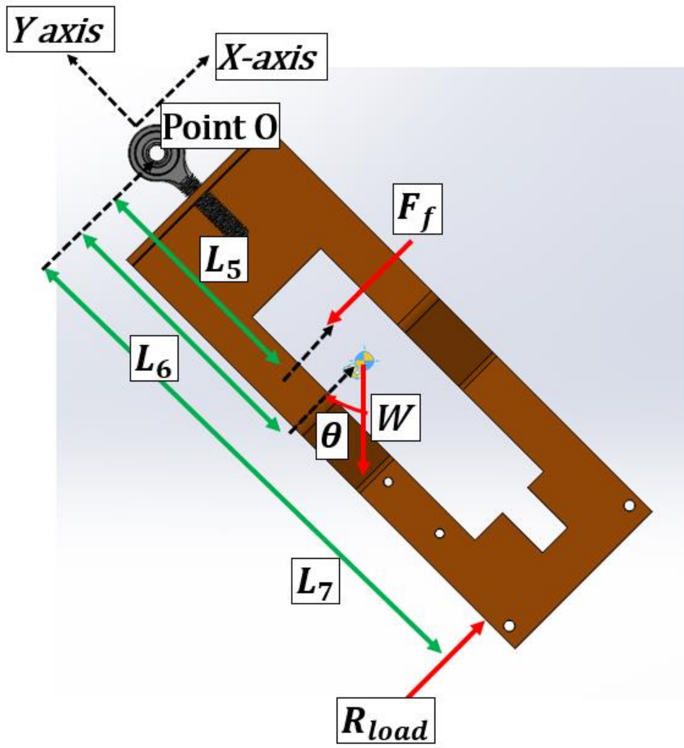
Assumptions:
1) Bike is stalled or there is little to no movement from yoke
2) Symmetry is assumed on either side of the flywheel



FBD of Flywheel

$$\sum M_P = 0 = -\tau_{flywheel} + F_f L_4 = 0$$

$$\tau_{flywheel} = F_f L_4$$



FBD of Brake Dyno

$$\sum M_O = 0 = -F_f L_5 - W\cos\theta\ L_6 + R_{\text{load}} L_7$$

$$F_f L_5 = R_{\text{load}} L_7 - W\cos\theta\ L_6$$

$$F_f = R_{\text{load}}\frac{L_7}{L_5} - W\cos\theta\ \frac{L_6}{L_5}$$

Substitution

$$\tau_{flywheel} = \left(R_{\text{load}}\frac{L_7}{L_5} - W\cos\theta\ \frac{L_6}{L_5}\right) L_4$$

Assuming weight will be offset

$$\tau_{flywheel} = R_{\text{load}}\frac{L_7 L_4}{L_5}$$

The load cell has a gain of 25 lbf/V and the voltage measured for weight was found to be 0.56 V. Therefore, the equation becomes:

$$\tau_{\text{flywheel}} = (V_{\text{load}} - 0.56\ \text{V})\left(25\ \frac{\text{lbf}}{\text{V}}\right)\frac{L_7 L_4}{L_5}$$

16 APPENDIX G: IR SENSOR RPS CALCULATION DERIVATION

Flywheel Circumference

$$C = d\pi$$

$$C = (460\ \text{mm})\pi = 1445.13\ \text{mm}$$

Strip Length (30 Strips were Chosen)

$$s = \frac{C}{2(30)} = \frac{d\pi}{60}$$

$$s = \frac{1445.13\ \text{mm}}{60} = 24.09\ \text{mm}$$

Angular Width of One Strip

$$\theta_{strip} = \frac{s}{r} = \frac{2s}{d}$$

$$\theta_{strip} = \frac{2(24.09)}{460} = 0.10474\ \text{rad}$$

Since each strip covered half the flywheel circumference each strip occupied

$$\theta_{strip} = \frac{\pi}{30} = 0.10472\ \text{rad}$$

Since the strips and the uncovered spaces were approximately equal in length the angular distance from the leading edge of one strip to the leading edge of the next strip is approximately twice the angular width of one strip.

$$\theta_{pulse} = 2\theta_{strip} = \frac{4s}{d}$$

$$\theta_{pulse} = 2(0.10472) = 0.20944 \text{ rad}$$

Therefore, the cumulative angular position at each detected edge is

$$\theta_n = Count_n \theta_{pulse}$$

Therefore, the angular velocity between two measurements is

$$\omega = \frac{\theta_n - \theta_{n-1}}{t_n - t_{n-1}}$$

The angular velocity was then converted to RPS using this equation

$$RPS = \frac{\omega}{2\pi}$$

$$RPS = \frac{1}{2\pi}\left(\frac{\theta_n - \theta_{n-1}}{t_n - t_{n-1}}\right)$$

Then the expression for the angular position at each edge detected was substituted in this expression.

$$RPS = \frac{\theta_{pulse}}{2\pi}\left(\frac{Count_n - Count_{n-1}}{t_n - t_{n-1}}\right)$$

Then the expression for theta pulse can be put into this equation

$$RPS = \frac{2\pi/30}{2\pi}\left(\frac{Count_n - Count_{n-1}}{t_n - t_{n-1}}\right)$$

Simplifying this expression gives

$$RPS = \frac{Count_n - Count_{n-1}}{30(t_n - t_{n-1})}$$

Since speed is calculated using two consecutive leading edges then

$$Count_n - Count_{n-1} = 1$$

Finally, the equation used for the estimation of the instantaneous velocity of the flywheel used is given below.

$$\boxed{RPS = \frac{1}{30(t_n - t_{n-1})}}$$

17 APPENDIX H: EXAMPLE DATA

Below is a table containing a cut data set from one of the tests.

Table 9. Cut Data Set Example

X(volts)	P(volts)
4.114565	1.001132
4.114565	1.00081
4.114565	1.000165
4.114887	1.000487

4.114887	1.00081
4.114243	1.002099
4.114887	1.001132
4.114565	1.002099
4.097157	1.394421
4.001736	2.169072
3.970144	2.390217
3.94242	2.282868
3.91792	2.023039
3.895354	2.173908
3.873111	2.116204
3.85119	2.068816
3.828624	2.103954
3.806381	2.098796
3.783493	2.08171
3.762539	2.059145
3.739328	2.068816
3.71773	2.029487
3.694841	2.035934
3.674855	2.022717
3.654868	2.030776
3.633591	2.031099
3.611993	2.046572
3.590072	2.058178
3.569118	2.054632
3.547519	2.062691
3.52592	2.065592
3.504644	2.091382
3.482401	2.080743
3.459835	2.089447
3.437914	2.108467
3.415993	2.086546
3.392782	2.082033
3.371828	2.072362
3.35023	2.074941
3.329598	2.068816
3.308967	2.064947
3.287368	2.073329
3.265769	2.056888
3.243848	2.054954
3.223861	2.059145
3.202585	2.051408

3.182276	2.05173
3.161644	2.054632
3.140046	2.037546
3.119092	2.051408
3.098782	2.036579
3.077828	2.044638
3.057197	2.050441
3.035276	2.037224
3.014967	2.061724
2.994335	2.061401
2.973059	2.063658
2.95146	2.046895
2.929539	2.044638
2.908585	2.030454
2.886986	2.020138
2.866677	2.020783
2.846046	1.98371
2.826059	1.958566
2.80446	1.949217
2.784151	1.908921
2.765131	1.895382
2.745789	1.878618
2.727092	1.856697
2.709039	1.845414
2.690342	1.830908
2.671644	1.82575
2.652302	1.798026
2.633605	1.781263
2.614908	1.765145
2.597177	1.752895
2.579447	1.732263
2.562039	1.728395
2.545276	1.714211
2.528835	1.716467
2.511105	1.697447
2.494342	1.686165
2.477579	1.683586
2.460171	1.676816
2.443085	1.674559
2.425355	1.662309
2.408914	1.644579
2.392796	1.642645

2.376355	1.622336
2.359914	1.621369
2.342506	1.615244
2.327033	1.604928
2.311881	1.595901
2.295763	1.601059
2.280289	1.589776
2.265138	1.588487
2.249342	1.574948
2.233546	1.57527
2.218072	1.558829
2.202599	1.532395
2.188092	1.506605
2.173585	1.462119
2.160046	1.427625
2.146506	1.381849
2.133612	1.345421
2.122329	1.30577
2.111691	1.271277
2.102342	1.241941
2.091704	1.216474
2.083322	1.192619
2.073651	1.168763
2.065592	1.143619
2.0585	1.1275
2.051408	1.110092
2.045605	1.091073
2.038835	1.074632
2.033355	1.059158
2.028842	1.046908

The figure below is the graph generated by the cut data.

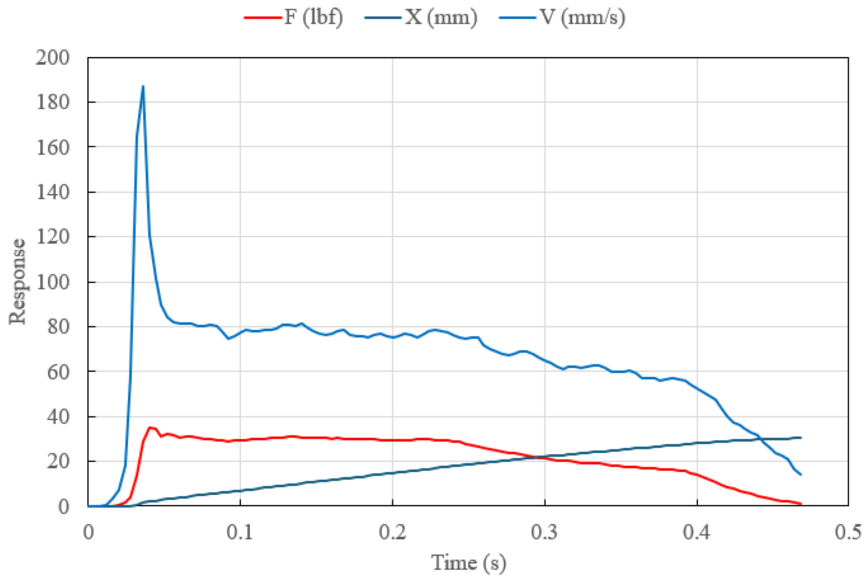


Figure 36. Example Graph

More data from tests is stored on flash drive and is available upon request.

18 APPENDIX I LOAD CELL DATASHEET

TE connectivity

✓RoHS CE

FC22

Compression Load Cell

SPECIFICATIONS

- **10 – 100 lbf Ranges**
- **High Level or mV Outputs**
- **Interchangeable**
- **Compact Easy to Fixture Design**
- **CE Compliance**

FEATURES

- Small Size
- Low Noise
- Robust: High Over-Range Capability
- High Reliability
- Low Deflection
- Essentially Unlimited Cycle Life Expectancy
- Low Off Center Errors
- Fast Response Time
- 10 to 100 lbf Ranges
- Reverse Polarity Protected

APPLICATIONS

- Medical Infusion Pumps
- Robotics End-Effectors
- Variable Force Control
- Load and Compression Sensing
- Exercise Machines
- Pumps
- Contact Sensing
- Weighing
- Household Appliances

The FC22 is a medium compression force sensor that creates new markets previously unrealizable due to cost and performance constraints. The FC22 offers normalized zero and span for interchangeability and is thermally compensated for changes in zero and span with respect to temperature.

The FC22 incorporates MEAS' proprietary Microfused technology which employs micromachined silicon piezoresistive strain gages fused with high temperature glass to a high performance stainless steel substrate. Microfused technology eliminates age-sensitive organic epoxies used in traditional load cell designs providing excellent long term span and zero stability. The FC22 measures direct force and is therefore not subject to lead-die fatigue failure common with competitive designs which use a pressure capsule embedded within a silicone gel-filled cavity. Operating at very low strains, Microfused technology provides an essentially unlimited cycle life expectancy, superior resolution, and high over-range capabilities.

Cost-optimization of the FC22 brings your OEM product to life whether you need thousands or millions of load cells annually. Although the standard model is ideal for a wide range of applications, our dedicated design team at our Load Cell Engineering Center is ready to provide you with custom designs for your OEM applications.

Please refer to the FS20 for lower force applications or the FC23 for higher force applications.

STANDARD RANGES

Range	lbf
0 to 0010	•
0 to 0025	•
0 to 0050	•
0 to 0100	•

PERFORMANCE SPECIFICATIONS

Supply Voltage: 5.0V, Ambient Temperature: 25°C (unless otherwise specified)

PARAMETERS	MIN	TYP	MAX	UNITS	NOTES
Span (Unamplified, FC2201)	24	30	36	mV/V	1
Span (Unamplified, FC2211)	19	20	21	mV/V	1
Span (Amplified)	3.8	4.00	4.2	V	1
Zero Force Output (Unamplified, FC2201)	-1	0	1	mV/V	1
Zero Force Output (Unamplified, FC2211)	-0.5	0	0.5	mV/V	1
Zero Force Output (Amplified)	0.45	0.5	0.55	V	1
Accuracy (non-linearity, hysteresis, and repeatability)		±1		%Span	2
Output Resistance (Unamplified)		2.2		kΩ	
Input Resistance (Unamplified)		3		kΩ	
Temperature Error – Zero	-1.25		1.25	%Span	3
Temperature Error – Span	-1.25		1.25	%Span	3
Long Term Stability (1 year)		±1		%Span	
Maximum Overload	2.5X			Rated	
Compensated Temperature	0		50	°C	
Operating Temperature	-40		+85	°C	
Storage Temperature	-40		+85	°C	
Excitation Voltage (Unamplified)			5	V_{DC}	
Excitation Voltage (Amplified)	4.75	5	5.25	V_{DC}	
Isolation Resistance ($250V_{dc}$)	50			MΩ	
Deflection at Rated Load			0.05	mm	
Humidity	0		90	%RH	
Weight		18.41		grams	

For custom configurations, consult factory

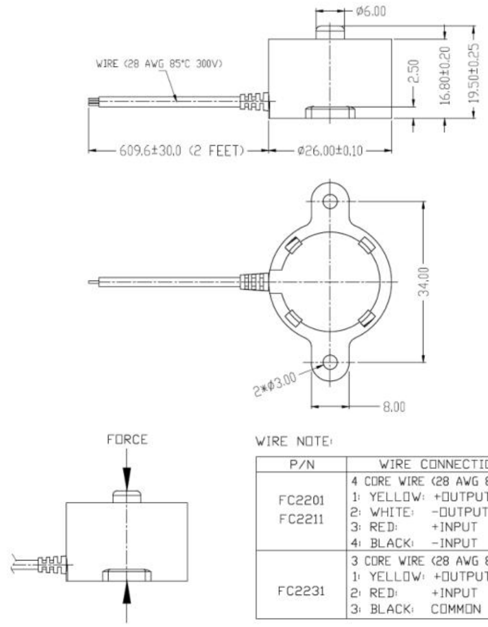
Notes
1. Ratiometric to supply.
2. Best fit straight line.
3. Maximum temperature error over compensated range with respect to 25°C.

CE Compliance

IEC61000-4-2 [4 kV/ 4kV (Air/Contact)]
IEC61000-4-3 (3 V/m)
IEC55022 Class A

DIMENSIONS

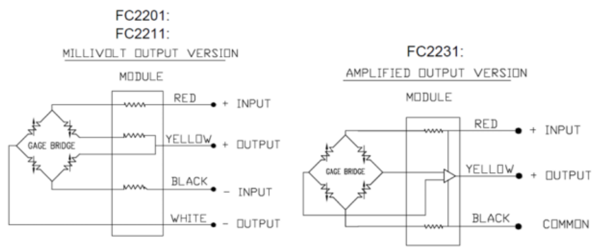
FC2201:
FC2211:
FC2231:



WIRE NOTE:

P/N	WIRE CONNECTIONS
FC2201 FC2211	4 CORE WIRE (28 AWG 85°C 300V) 1: YELLOW: +OUTPUT 2: WHITE: -OUTPUT 3: RED: +INPUT 4: BLACK: -INPUT
FC2231	3 CORE WIRE (28 AWG 85°C 300V) 1: YELLOW: +OUTPUT 2: RED: +INPUT 3: BLACK: COMMON

CONNECTIONS



ORDERING INFORMATION

FC22 – 3 1 0000 0100 – L

Output	
Code	Output
0	0 – 150 mV
1	0 – 100 mV
3	0.5 – 4.5 V

Connection	
1	2 feet Cable

Vent	
L	lbf

Multiplier	
-	None

Force Range
0010
0025
0050
0100

NORTH AMERICA
Measurement Specialties, Inc.,
a TE Connectivity Company
Tel: 1-800-522-6752
Email : customercare.lcsb@te.com

EUROPE
MEAS France SAS,
a TE Connectivity company
Tel: +31 73 624 6999
Email : customercare.lcsb@te.com

ASIA
Measurement Specialties (China), Ltd.,
a TE Connectivity Company
Tel: +86 400-820-6015
Email: customercare.shzn@te.com